Ordinary Differential Equations

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Introduction

An ordinary differential equation describes a relation between an unknown function of one variable and its derivatives. At first sight, the main task of ODE theory seems to be straightforward: given an equation, find its solution. For a small collection of equations this can indeed be done explicitly, using techniques such as separation of variables, integrating factors, elementary transformations, or the method of variation of parameters. However, explicit formulas form only one part of the theory. Many differential equations cannot be solved in elementary form, and even when a formula exists, it may not reveal the most important features of the solution. This leads naturally to several deeper questions: does a solution exist, is it unique, how long can it be continued, and what happens to it as time evolves?

These questions suggest three closely related viewpoints for studying ordinary differential equations:

$$\text{explicit solution} \longrightarrow \text{existence and uniqueness} \longrightarrow \text{qualitative behavior}.$$

The first viewpoint develops methods for solving important classes of equations. The second establishes a general theory that guarantees the existence, uniqueness, and continuation of solutions even when no closed form is available. The third regards an ODE as a dynamical system and studies trajectories, equilibria, stability, periodic orbits, and long-time behavior.

Linear equations provide the central bridge between these viewpoints. For higher-order linear equations, the solution set forms a finite dimensional vector space, and concepts such as linear independence, Wronskians, characteristic roots, and variation of parameters arise naturally. A higher-order equation can also be rewritten as a system of first-order equations, which leads to fundamental matrices and the matrix exponential.

For nonlinear autonomous systems, explicit solution formulas are usually unavailable. Instead, the geometry of trajectories becomes the main object of study. Linearization describes the local behavior near an equilibrium when it is sufficiently nondegenerate, while Lyapunov functions give a nonlinear method for proving stability. Periodic solutions and limit cycles lead further to planar dynamical systems and the Poincaré–Bendixson theory.

The final part of these notes concerns Sturm–Liouville boundary value problems. Here the viewpoint changes from initial values to boundary conditions and from solving a single equation to studying an eigenvalue problem for a differential operator. The resulting orthogonality and eigenfunction structure form an important connection between ordinary differential equations, Fourier analysis, and partial differential equations.

The purpose of these notes is therefore not merely to collect solution formulas. The aim is to organize the subject around the progression from computation to theory and then to dynamics, while preserving the main conclusions, proofs, examples, and qualitative insights developed throughout the course.

Chapter 1

First-Order Differential Equations

1.1 Basic Concepts and Direction Fields

An ordinary differential equation, abbreviated as an ODE, is an equation involving an unknown function of one independent variable and its derivatives. A general n-th order ODE may be written as

$$F\Big(t, y(t), y'(t), \ldots, y^{(n)}(t)\Big) = 0.$$

When the highest derivative can be isolated, it may also be written in normal form as

$$y^{(n)} = f\Big(t, y, y', \ldots, y^{(n-1)}\Big).$$

Definition 1.1.1. The *order* of an ordinary differential equation is the order of the highest derivative appearing in the equation.

An n-th order equation is called *linear* if it has the form

$$a_0(t)y^{(n)} + a_1(t)y^{(n-1)} + \cdots + a_{n-1}(t)y' + a_n(t)y = g(t),$$

where the coefficients $a_0, \ldots, a_n$ depend only on the independent variable t. Otherwise, the equation is nonlinear.
For a first-order equation

$$y' = f(t, y),$$

an initial condition specifies the value of the solution at some initial point t_0:

$$y(t_0) = y_0.$$

The resulting problem

$$\begin{cases} y' = f(t, y), \\ y(t_0) = y_0 \end{cases}$$

is called an *initial value problem*, or IVP.
A family of solutions containing arbitrary constants is called a *general solution*. Once the constants are determined by initial conditions, the corresponding member of the family is a *particular solution*.

Definition 1.1.2. For the equation $y' = f(t, y)$, a *direction field* or *slope field* assigns to each point (t, y) a small line segment having slope $f(t, y)$.

The direction field gives qualitative information about the solution curves even before the differential equation is solved explicitly.

Example 1.1.3. Consider

$$y' = ay - b, \qquad y(0) = y_0,$$

where a and b are constants.
If $a \neq 0$, then

$$y' - ay = -b.$$

The equilibrium value is b/a. Writing $z = y - b/a$, we obtain $z' = az$, so

$$z(t) = Ce^{at}.$$

Hence

$$y(t) = \frac{b}{a} + \left(y_0 - \frac{b}{a}\right) e^{at}.$$

If $a = 0$, then the equation reduces to $y' = -b$, and therefore

$$y(t) = y_0 - bt.$$

1.2 First-Order Linear Equations

A first-order linear equation has the standard form

$$y' + p(t)y = g(t).$$

Definition 1.2.1. A nonzero function $\mu(t)$ is called an *integrating factor* if multiplication by $\mu(t)$ transforms the left-hand side into a derivative:

$$\mu(t)y' + \mu(t)p(t)y = \frac{d}{dt}\Big(\mu(t)y(t)\Big).$$

For this identity to hold, we need

$$\mu'(t) = p(t)\mu(t).$$

Thus we may take

$$\mu(t) = \exp\left(\int p(t)\,dt\right).$$

Multiplying the differential equation by $\mu(t)$ gives

$$\frac{d}{dt}\Big(\mu(t)y(t)\Big) = \mu(t)g(t).$$

Integrating,

$$\mu(t)y(t) = \int \mu(t)g(t)\,dt + C,$$

and hence

$$y(t) = \frac{1}{\mu(t)}\left[\int \mu(t)g(t)\,dt + C\right].$$

For an initial value $y(t_0) = y_0$, it is convenient to write

$$\mu(t) = \exp\left(\int_{t_0}^{t} p(s)\,ds\right).$$

Then

$$y(t) = e^{-\int_{t_0}^{t} p(s)\,ds}\left[y_0 + \int_{t_0}^{t} e^{\int_{t_0}^{s} p(r)\,dr} g(s)\,ds\right].$$

Remark 1.2.2. The integrating-factor method is not an isolated trick. It is designed precisely so that the two terms involving y combine into the derivative of a product.

1.3 Variation of Parameters

The same formula for a first-order linear equation can be derived from the homogeneous solution.
Consider first

$$y' + p(t)y = 0.$$

Its general solution is

$$y_h(t) = Ce^{-\int p(t)\,dt}.$$

Variation of parameters replaces the constant C by an unknown function $a(t)$:

$$y(t) = a(t)e^{-\int p(t)\,dt}.$$

Differentiating,

$$y' = a'(t)e^{-\int p(t)\,dt} - p(t)a(t)e^{-\int p(t)\,dt}.$$

Therefore

$$y' + p(t)y = a'(t)e^{-\int p(t)\,dt}.$$

For the nonhomogeneous equation

$$y' + p(t)y = g(t),$$

we obtain

$$a'(t) = g(t)e^{\int p(t)\,dt}.$$

Thus

$$a(t) = \int g(t)e^{\int p(t)\,dt}\,dt + C,$$

and consequently

$$y(t) = e^{-\int p(t)\,dt}\left[\int g(t)e^{\int p(t)\,dt}\,dt + C\right].$$

Remark 1.3.1. For first-order linear equations, the integrating-factor method and variation of parameters lead to exactly the same solution formula. The difference lies only in the viewpoint: the former searches for a product derivative, while the latter varies the constant in the homogeneous solution.

1.4 Separable Differential Equations

A first-order equation is called separable if it can be written as

$$y' = f(t)g(y).$$

Before dividing by $g(y)$, one should first solve

$$g(y) = 0.$$

If $g(a) = 0$, then $y(t) = a$ is a constant solution.
Assume now that $g(y) \neq 0$. Then

$$\frac{dy}{g(y)} = f(t)\,dt.$$

Integrating both sides gives

$$\int \frac{1}{g(y)}\,dy = \int f(t)\,dt + C.$$

The resulting relation may define y explicitly or only implicitly.
More generally, an equation of the form

$$M(x)\,dx + N(y)\,dy = 0$$

is immediately separable:

$$\int M(x)\,dx + \int N(y)\,dy = C.$$

Remark 1.4.1. Dividing by $g(y)$ may remove constant solutions satisfying $g(y) = 0$. Therefore these solutions must always be checked before performing the division.

1.5 Exact Differential Equations

Consider

$$M(x,y)\,dx + N(x,y)\,dy = 0.$$

Definition 1.5.1. The equation is called *exact* on a domain D if there exists a scalar function $\Phi(x,y)$ such that

$$\Phi_x(x,y) = M(x,y), \qquad \Phi_y(x,y) = N(x,y).$$

If $y = y(x)$ is a solution, then

$$\frac{d}{dx}\Phi(x,y(x)) = \Phi_x + \Phi_y y' = M + Ny' = 0.$$

Therefore

$$\Phi(x,y) = C,$$

so the solution curves are level curves of the potential function Φ.

Theorem 1.5.2. Suppose M, N, M_y, and N_x are continuous on a rectangular domain D. Then

$$M(x,y)\,dx + N(x,y)\,dy = 0$$

is exact on D if and only if

$$M_y = N_x$$

throughout D.

Proof. Suppose first that the equation is exact. Then there exists Φ such that

$$\Phi_x = M, \qquad \Phi_y = N.$$

By equality of mixed partial derivatives,

$$M_y = \Phi_{xy} = \Phi_{yx} = N_x.$$

Conversely, assume $M_y = N_x$. Fix $(x_0, y_0) \in D$ and define

$$\Phi(x,y) = \int_{x_0}^{x} M(s,y)\,ds + h(y),$$

where h is to be determined. Then

$$\Phi_x(x,y) = M(x,y).$$

Also,

$$\begin{aligned}\Phi_y(x,y) &= \int_{x_0}^{x} M_y(s,y)\,ds + h'(y)\\ &= \int_{x_0}^{x} N_x(s,y)\,ds + h'(y)\\ &= N(x,y) - N(x_0,y) + h'(y).\end{aligned}$$

Choose

$$h'(y) = N(x_0,y).$$

Then $\Phi_y = N$, so the equation is exact. □

Example 1.5.3. Consider

$$(2t + y^2)\,dt + 2ty\,dy = 0.$$

Here

$$M(t, y) = 2t + y^2, \qquad N(t, y) = 2ty.$$

Since

$$M_y = 2y = N_t,$$

the equation is exact.
We seek Φ satisfying

$$\Phi_t = 2t + y^2.$$

Integrating with respect to t,

$$\Phi(t, y) = t^2 + ty^2 + h(y).$$

Then

$$\Phi_y = 2ty + h'(y).$$

Comparing with $N = 2ty$, we obtain $h'(y) = 0$. Thus we may take

$$\Phi(t, y) = t^2 + ty^2.$$

Hence the solutions are determined implicitly by

$$t^2 + ty^2 = C.$$

1.6 Integrating Factors for Nonexact Equations

A differential equation

$$M(x, y)\,dx + N(x, y)\,dy = 0$$

may fail to be exact. In some cases, multiplication by a nonzero function $\mu(x, y)$ makes it exact:

$$\mu M\,dx + \mu N\,dy = 0.$$

Definition 1.6.1. Such a function μ is called an *integrating factor*.

The exactness condition becomes

$$\frac{\partial}{\partial y}(\mu M) = \frac{\partial}{\partial x}(\mu N).$$

Equivalently,

$$M\mu_y - N\mu_x = (N_x - M_y)\mu.$$

In general, finding $\mu(x, y)$ may be difficult. Two useful special cases occur when the integrating factor depends only on one variable.
If $\mu = \mu(x)$, then

$$\frac{\mu'}{\mu} = \frac{M_y - N_x}{N},$$

provided the right-hand side depends only on x.
If $\mu = \mu(y)$, then

$$\frac{\mu'}{\mu} = \frac{N_x - M_y}{M},$$

provided the right-hand side depends only on y.

1.7 Homogeneous First-Order Equations

Definition 1.7.1. A first-order equation

$$y' = f(x, y)$$

is called homogeneous in this context if the right-hand side depends only on the ratio y/x:

$$f(x, y) = g\left(\frac{y}{x}\right).$$

This meaning of "homogeneous" is different from the term *homogeneous linear equation*, where the nonhomogeneous term is zero.
Set

$$u = \frac{y}{x}, \qquad y = xu.$$

Then

$$y' = u + xu'.$$

Hence

$$u + xu' = g(u),$$

so

$$xu' = g(u) - u.$$

Whenever $g(u) - u \neq 0$, this becomes the separable equation

$$\frac{du}{g(u) - u} = \frac{dx}{x}.$$

Thus a homogeneous first-order equation is reduced to a separable equation by the substitution $y = xu$.

1.8 Equations Reducible by Translation

Some equations have the form

$$\frac{dy}{dx} = f\left(\frac{ax + by + c}{mx + ny + \ell}\right),$$

where a, b, c, m, n, ℓ are constants.
When

$$an - bm \neq 0,$$

we may choose constants x_0, y_0 satisfying

$$ax_0 + by_0 + c = 0, \qquad mx_0 + ny_0 + \ell = 0.$$

Set

$$X = x - x_0, \qquad Y = y - y_0.$$

Then the constant terms disappear and the equation becomes

$$\frac{dY}{dX} = f\left(\frac{aX + bY}{mX + nY}\right),$$

which is homogeneous and can be treated using $Y = uX$.

If

$$an - bm = 0,$$

the numerator and denominator involve the same linear combination of x and y. In that case, a substitution such as

$$v = ax + by$$

reduces the equation to a separable equation.

1.9 Bernoulli Equations

A Bernoulli equation has the form

$$y' + p(x)y = q(x)y^n, \qquad n \neq 0, 1.$$

Multiplying by $(1-n)y^{-n}$ gives

$$(1-n)y^{-n}y' + (1-n)p(x)y^{1-n} = (1-n)q(x).$$

Set

$$z = y^{1-n}.$$

Then

$$z' = (1-n)y^{-n}y',$$

and the equation becomes

$$z' + (1-n)p(x)z = (1-n)q(x),$$

which is a first-order linear equation.

Remark 1.9.1. The Bernoulli substitution illustrates a recurring principle in ODEs: a nonlinear equation may become linear after a suitable change of dependent variable.

1.10 Riccati Equations

A Riccati equation has the form

$$y' = p(x)y^2 + q(x)y + r(x).$$

Several special cases are immediately simpler. If $p(x) = 0$, the equation is linear. If $r(x) = 0$, it is a Bernoulli equation with exponent 2.
The general Riccati equation is nonlinear, but knowledge of one particular solution reduces it to a first-order linear equation.

Theorem 1.10.1. Suppose

$$y' = p(x)y^2 + q(x)y + r(x)$$

has a particular solution $y = \phi(x)$. Then its general solution can be reduced to the solution of a first-order linear equation.

Proof. Set

$$y = \phi + u.$$

Since

$$\phi' = p\phi^2 + q\phi + r,$$

substitution gives

$$u' = \Big(2p\phi + q\Big)u + pu^2.$$

For $u \neq 0$, let $z = u^{-1}$. Then

$$z' = -\frac{u'}{u^2},$$

so

$$z' + \Big(2p\phi + q\Big)z = -p.$$

This is a first-order linear differential equation for z. □

Remark 1.10.2. The important point is not that every Riccati equation can be solved explicitly. Rather, one known particular solution changes the problem from nonlinear to linear.

1.11 Modeling with First-Order Equations

First-order differential equations arise naturally when the rate of change of a quantity is determined by its current state.

Model 1.11.1 (Continuous Compound Interest)**.** Let $S(t)$ denote the balance of an account at time t, and let r be the continuously compounded annual interest rate. Then

$$S' = rS, \qquad S(0) = S_0.$$

The solution is

$$S(t) = S_0 e^{rt}.$$

If money is deposited continuously at a constant rate K, then

$$S' = rS + K, \qquad S(0) = S_0.$$

For $r \neq 0$,

$$S(t) = S_0 e^{rt} + \frac{K}{r}\Big(e^{rt} - 1\Big).$$

Model 1.11.2 (Mortgage)**.** Suppose $S(t)$ is the remaining balance of a loan, r is the continuous interest rate, and payments are made continuously at rate P. Then

$$S' = rS - P.$$

With $S(0) = S_0$,

$$S(t) = S_0 e^{rt} - \frac{P}{r}\Big(e^{rt} - 1\Big).$$

Model 1.11.3 (Logistic Growth)**.** Let $P(t)$ denote a population. The logistic model is

$$P' = rP\left(1 - \frac{P}{K}\right),$$

where $r > 0$ is the intrinsic growth rate and $K > 0$ is the environmental carrying capacity.

The equilibrium solutions are

$$P = 0 \qquad \text{and} \qquad P = K.$$

For $0 < P < K$, the population increases, while for $P > K$, it decreases toward K. This simple model already illustrates why qualitative information may be more useful than an explicit solution.

Model 1.11.4 (Predator–Prey System)**.** Let $x(t)$ denote the prey population and $y(t)$ the predator population. The Lotka–Volterra model is

$$\begin{cases} x' = x(r - by), \\ y' = y(-s + cx), \end{cases}$$

where $r, b, s, c > 0$.
Here r is the growth rate per capita of the prey in the absence of predators, b measures the decrease caused by predators, s is the death rate per capita of predators in the absence of prey, and c measures the increase caused by the prey population.
For $x, y > 0$,

$$\frac{dy}{dx} = \frac{y(-s + cx)}{x(r - by)}.$$

Separating variables gives

$$\left(\frac{r}{y} - b\right) dy = \left(c - \frac{s}{x}\right) dx.$$

Hence

$$r \ln y - by = cx - s \ln x + C.$$

Equivalently,

$$V(x, y) = s \ln x - cx + r \ln y - by$$

is constant along every positive solution trajectory.
Indeed,

$$\begin{aligned} \frac{d}{dt} V(x(t), y(t)) &= \left(\frac{s}{x} - c\right) x' + \left(\frac{r}{y} - b\right) y' \\ &= (s - cx)(r - by) + (r - by)(-s + cx) \\ &= 0. \end{aligned}$$

Thus V is a constant of motion.
The positive equilibrium is

$$\left(\frac{s}{c}, \frac{r}{b}\right).$$

The closed level curves surrounding this equilibrium already suggest the periodic behavior that will later be studied systematically using the language of autonomous systems and phase portraits.

Remark 1.11.5. The examples above illustrate two different roles of differential equations. In simple models such as compound interest, an explicit formula gives nearly all relevant information. In nonlinear models such as the predator–prey system, the geometry and qualitative behavior of solution curves become equally important. This transition from explicit solution methods to qualitative theory motivates the next chapter.

Chapter 2

Qualitative Theory of First-Order ODEs

2.1 Autonomous Equations and Phase Lines

A first-order differential equation is called *autonomous* if it has the form

$$y' = f(y),$$

where the independent variable does not appear explicitly.

Definition 2.1.1. A constant solution $y(t) \equiv y_*$ is called an *equilibrium solution* if

$$f(y_*) = 0.$$

Even when the equation cannot be solved explicitly, the sign of $f(y)$ determines whether solutions increase or decrease. This leads to the phase-line method.
Suppose the equilibrium points are

$$y_1 < y_2 < \cdots < y_k.$$

On every interval between consecutive equilibria, $f(y)$ has a fixed sign unless it vanishes again. Thus

$$f(y) > 0 \quad \Longrightarrow \quad y'(t) > 0,$$

so solutions move toward larger values of y, while

$$f(y) < 0 \quad \Longrightarrow \quad y'(t) < 0,$$

so solutions move toward smaller values of y.

Definition 2.1.2. An equilibrium y_* is called

- *stable* if nearby solutions remain near y_*;
- *asymptotically stable* if it is stable and nearby solutions converge to y_*;
- *unstable* if it is not stable.

In a one-dimensional autonomous equation, the stability of an isolated equilibrium can usually be read directly from the arrows on the phase line.
If the arrows point toward y_* from both sides, then y_* is asymptotically stable. If the arrows point away from y_* on both sides, then it is unstable.
A simple local criterion is obtained by differentiation.

Proposition 2.1.3. Suppose $f \in C^1$ near an equilibrium y_*.
If

$$f'(y_*) < 0,$$

then y_* is locally asymptotically stable.
If

$$f'(y_*) > 0,$$

then y_* is unstable.

Remark 2.1.4. When $f'(y_*) = 0$, the linear approximation does not decide the stability. The sign of $f(y)$ itself must then be examined more carefully.

2.2 The Logistic Equation

The logistic model is

$$P' = rP\left(1 - \frac{P}{K}\right),$$

where $r > 0$ is the intrinsic growth rate and $K > 0$ is the carrying capacity.
The equilibrium points satisfy

$$rP\left(1 - \frac{P}{K}\right) = 0,$$

so

$$P = 0 \qquad \text{or} \qquad P = K.$$

For $0 < P < K$,

$$P' > 0,$$

while for $P > K$,

$$P' < 0.$$

Therefore $P = K$ is asymptotically stable, whereas $P = 0$ is unstable when only positive populations are considered.
The logistic equation is explicitly solvable, but its phase line already reveals its essential long-time behavior:

$$P(t) \longrightarrow K$$

for every positive initial population.

Remark 2.2.1. This example illustrates one of the central ideas of qualitative ODE theory: an explicit formula is often unnecessary if the main goal is to determine monotonicity, equilibria, stability, or limiting behavior.

2.3 Bifurcation

Many differential equations contain a parameter that represents a physical or mathematical quantity. Consider

$$y' = f(y, c),$$

where $c \in \mathbb{R}$ is a parameter.

Definition 2.3.1. A *bifurcation* occurs when the qualitative structure of the equilibria changes as the parameter varies. In particular, equilibrium points may appear, disappear, merge, or change their stability.

A simple example is

$$y' = y^2 - c.$$

The equilibrium equation is

$$y^2 - c = 0.$$

If $c < 0$, there are no real equilibrium points. If $c = 0$, there is one equilibrium $y = 0$. If $c > 0$, there are two equilibria

$$y = \pm\sqrt{c}.$$

Thus $c = 0$ is a bifurcation value.
This is the standard geometry of a saddle-node bifurcation.

2.4 Logistic Growth with Harvesting

Consider the logistic equation with constant harvesting:

$$P' = rP\left(1 - \frac{P}{K}\right) - H,$$

where $H \geq 0$ is the harvesting rate.
Define

$$g(P) = rP\left(1 - \frac{P}{K}\right).$$

Equilibria satisfy

$$g(P) = H.$$

Since $g(P)$ is a concave quadratic function, its maximum occurs at

$$P = \frac{K}{2}.$$

The maximal sustainable harvesting rate is therefore

$$H_0 = g\left(\frac{K}{2}\right) = \frac{rK}{4}.$$

Hence:

$$\begin{cases} H < H_0, & \text{there are two positive equilibria,} \\ H = H_0, & \text{there is one positive equilibrium,} \\ H > H_0, & \text{there is no positive equilibrium.} \end{cases}$$

Thus

$$H = H_0$$

is a bifurcation value.
The two equilibria for $H < H_0$ are

$$P_{\pm} = \frac{K}{2}\left(1 \pm \sqrt{1 - \frac{4H}{rK}}\right).$$

The smaller equilibrium is unstable, while the larger equilibrium is asymptotically stable.

Remark 2.4.1. The bifurcation has a direct modeling interpretation. If harvesting exceeds H_0, the logistic growth term can never compensate for the removal rate, so no positive equilibrium population remains.

2.5 Pitchfork-Type Bifurcation

Consider

$$y' = (c - y^2)y.$$

The equilibria satisfy

$$(c - y^2)y = 0.$$

Thus $y = 0$ is always an equilibrium, while

$$y = \pm\sqrt{c}$$

appear when $c > 0$.
For $c < 0$, only $y = 0$ exists. For $c > 0$, there are three equilibria:

$$0, \qquad -\sqrt{c}, \qquad \sqrt{c}.$$

At $c = 0$, the equilibrium structure changes, so $c = 0$ is a bifurcation value. The corresponding bifurcation diagram has the characteristic pitchfork shape.

2.6 Existence and Uniqueness

We now turn from methods of solving equations to the fundamental question of whether a solution exists at all.
Consider the initial value problem

$$\begin{cases} y' = f(x, y), \\ y(x_0) = y_0. \end{cases}$$

Let

$$R = \{(x, y) : |x - x_0| \leq a, \quad |y - y_0| \leq b\}.$$

Definition 2.6.1. The function f is called *Lipschitz continuous with respect to* y on R if there exists $L > 0$ such that

$$|f(x, y_1) - f(x, y_2)| \leq L\,|y_1 - y_2|$$

for all $(x, y_1), (x, y_2) \in R$.

Theorem 2.6.2 (Existence and Uniqueness Theorem)**.** Suppose f is continuous on R and Lipschitz continuous with respect to y on R. Let

$$M = \max_{(x,y)\in R} |f(x, y)|$$

and

$$h = \min\left\{a, \frac{b}{M}\right\}.$$

Then the initial value problem

$$\begin{cases} y' = f(x, y), \\ y(x_0) = y_0 \end{cases}$$

has a unique solution on

$$I = [x_0 - h, x_0 + h].$$

The theorem is also referred to as the Picard existence and uniqueness theorem.

2.7 The Integral Equation Formulation

The differential equation

$$y' = f(x, y), \qquad y(x_0) = y_0$$

is equivalent to the integral equation

$$y(x) = y_0 + \int_{x_0}^{x} f(t, y(t))\,dt.$$

Indeed, integrating $y' = f(x, y)$ from x_0 to x gives the integral equation. Conversely, if y satisfies the integral equation and f is continuous, differentiation gives

$$y'(x) = f(x, y(x)).$$

This observation is the starting point of Picard iteration.

2.8 Picard Iteration

Define the first approximation by

$$y_0(x) = y_0.$$

Then recursively define

$$y_{n+1}(x) = y_0 + \int_{x_0}^{x} f(t, y_n(t))\, dt.$$

Thus

$$y_1(x) = y_0 + \int_{x_0}^{x} f(t, y_0)\, dt,$$

and

$$y_2(x) = y_0 + \int_{x_0}^{x} f(t, y_1(t))\, dt,$$

and so on.
The choice

$$h = \min\left\{a, \frac{b}{M}\right\}$$

ensures that every Picard iterate remains inside the rectangle R.
Indeed, if $|x - x_0| \le h$, then

$$\begin{aligned} |y_{n+1}(x) - y_0| &\le \int_{x_0}^{x} |f(t, y_n(t))|\, dt \\ &\le M\,|x - x_0| \\ &\le Mh \\ &\le b. \end{aligned}$$

2.9 Convergence of Picard Iteration

The Lipschitz condition controls the difference between successive iterations. We first have

$$|y_1(x) - y_0| \le M\,|x - x_0|\,.$$

Then

$$\begin{aligned} |y_2(x) - y_1(x)| &\le \int_{x_0}^{x} |f(t, y_1(t)) - f(t, y_0(t))|\, dt \\ &\le L \int_{x_0}^{x} |y_1(t) - y_0(t)|\, dt \\ &\le ML\frac{|x - x_0|^2}{2!}. \end{aligned}$$

Repeating this estimate gives

$$|y_{n+1}(x) - y_n(x)| \le \frac{ML^n\,|x - x_0|^{n+1}}{(n+1)!}.$$

Therefore

$$\sum_{n=0}^{\infty} |y_{n+1}(x) - y_n(x)|$$

is dominated by a convergent exponential series. Hence $\{y_n\}$ converges uniformly on I.
Let

$$y(x) = \lim_{n\to\infty} y_n(x).$$

Passing to the limit in

$$y_{n+1}(x) = y_0 + \int_{x_0}^{x} f(t, y_n(t))\, dt$$

gives

$$y(x) = y_0 + \int_{x_0}^{x} f(t, y(t))\, dt.$$

Thus y solves the IVP.

2.10 Uniqueness of the Solution

Suppose y and z are two solutions of the same IVP. Then

$$y(x) - z(x) = \int_{x_0}^{x} [f(t, y(t)) - f(t, z(t))]\, dt.$$

Hence

$$|y(x) - z(x)| \leq L \int_{x_0}^{x} |y(t) - z(t)|\ dt.$$

Iterating the estimate gives

$$|y(x) - z(x)| \leq \frac{L^n |x - x_0|^n}{n!} \max_{t \in I} |y(t) - z(t)|\,.$$

As $n \to \infty$,

$$\frac{L^n |x - x_0|^n}{n!} \longrightarrow 0.$$

Therefore

$$y(x) = z(x),$$

and the solution is unique.

2.11 Local Lipschitz Conditions

For extension arguments, it is useful to formulate the Lipschitz condition locally.

Definition 2.11.1. Let $D \subset \mathbb{R}^2$ be a domain. We say that $f(x, y)$ is *locally Lipschitz continuous with respect to* y on D if, for every bounded closed rectangle $R \subset D$, there exists a constant $L_R > 0$ such that

$$|f(x, y_1) - f(x, y_2)| \leq L_R\, |y_1 - y_2|$$

whenever $(x, y_1), (x, y_2) \in R$.

A convenient sufficient condition is differentiability with respect to y.

Proposition 2.11.2. Suppose f_y is continuous on a rectangle R. Then f is Lipschitz continuous with respect to y on R.

Proof. Since R is compact and f_y is continuous,

$$L = \max_{R} |f_y| < \infty.$$

For fixed x, the mean value theorem gives

$$f(x, y_1) - f(x, y_2) = f_y(x, \xi)(y_1 - y_2)$$

for some ξ between y_1 and y_2. Hence

$$|f(x, y_1) - f(x, y_2)| \leq L\, |y_1 - y_2|\,.$$

□

2.12 Uniqueness Beyond the Lipschitz Condition

The Lipschitz condition is a sufficient condition for uniqueness, but it is not necessary. A more general uniqueness criterion can be formulated in terms of a function that controls the difference

$$|f(x, y_1) - f(x, y_2)|.$$

Remark 2.12.1 (Osgood Condition)**.** Suppose

$$|f(x, y_1) - f(x, y_2)| \leq F\Big(|y_1 - y_2|\Big),$$

where $F(r) > 0$ for $r > 0$, F is continuous, and

$$\int_{0^+} \frac{dr}{F(r)} = +\infty.$$

Then uniqueness may still be obtained.
The Lipschitz condition corresponds to the special choice

$$F(r) = Lr.$$

Thus Lipschitz continuity is sufficient for uniqueness, but is not the most general possible condition.

2.13 Failure of Uniqueness

Continuity of f alone generally guarantees local existence, but does not guarantee uniqueness.

Example 2.13.1. Consider

$$\begin{cases} y' = 3y^{2/3}, \\ y(0) = 0. \end{cases}$$

One solution is

$$y(t) \equiv 0.$$

Another solution is

$$y(t) = t^3.$$

More generally, for any $a \geq 0$,

$$y(t) = \begin{cases} 0, & t \leq a, \\ (t-a)^3, & t > a \end{cases}$$

is also a solution after a corresponding translation of the initial point.
The function $f(y) = 3y^{2/3}$ is continuous but is not locally Lipschitz near $y = 0$. Therefore the existence theorem applies, but the usual uniqueness hypothesis fails.

2.14 Maximal and Saturated Intervals

The local existence theorem provides a solution only on a possibly small interval around the initial point. A natural question is how far that solution can be continued.

Definition 2.14.1. An interval $I_{\max}$ is called the *maximal interval of existence* of a solution if the solution exists on $I_{\max}$ and cannot be extended as a solution to any strictly larger interval.

The original notes also refer to such an interval as a *saturated interval*.
For a locally Lipschitz differential equation, uniqueness implies that local solutions obtained at different points must agree wherever their domains overlap. Therefore they can be joined to form a unique maximal solution.

2.15 Extension of Solutions

Consider

$$\begin{cases} y' = f(x, y), \\ y(x_0) = y_0, \end{cases}$$

where f is continuous on a domain $D \subset \mathbb{R}^2$.

Theorem 2.15.1 (Extension Theorem)**.** Every solution curve through $(x_0, y_0) \in D$ can be extended in both directions until it approaches the boundary of D, provided the hypotheses required for local existence continue to hold.

If f is also locally Lipschitz with respect to y, then the extension is unique.
Geometrically, the solution cannot simply stop at an interior point of the domain. If the graph remains inside a compact subset of D, the local existence theorem can be applied again at points near the end of the interval, producing a further extension.
Therefore, if a maximal solution ceases to exist at a finite endpoint, its graph must eventually leave every compact subset of the domain.

2.16 Finite-Time Blow-Up

The extension theorem explains how a solution may fail to exist globally.

Example 2.16.1. Consider

$$\begin{cases} y' = y^2, \\ y(0) = 1. \end{cases}$$

Separating variables gives

$$-\frac{1}{y} = t + C.$$

Using $y(0) = 1$,

$$y(t) = \frac{1}{1 - t}.$$

Thus the maximal interval is

$$I_{\max} = (-\infty, 1).$$

As $t \to 1^-$,

$$y(t) \longrightarrow +\infty.$$

The differential equation itself is smooth on all of $\mathbb{R}^2$, but the solution becomes unbounded in finite time. Hence smoothness of the vector field alone does not imply global existence for nonlinear equations.

2.17 A Riccati Example

Consider

$$\begin{cases} y' = x + y^2, \\ y(0) = 0. \end{cases}$$

The right-hand side

$$f(x, y) = x + y^2$$

is continuously differentiable on $\mathbb{R}^2$, so local existence and uniqueness follow immediately. However, the local theorem does not determine the entire maximal interval of the solution. In particular, a nonlinear solution may become unbounded in finite time even though f is smooth everywhere.

Remark 2.17.1. This distinction is fundamental:

$$\text{local existence} \neq \text{global existence}.$$

The Picard theorem guarantees a local solution, while the extension theorem and additional a priori estimates are needed to determine how far that solution continues.

2.18 Comparison Principles

Comparison methods allow one solution to be bounded by another without solving either equation exactly.
Suppose

$$\phi'(x) \leq f(x, \phi(x)),$$

and

$$\psi'(x) = f(x, \psi(x)).$$

Under suitable regularity assumptions, if

$$\phi(x_0) \leq \psi(x_0),$$

then one expects

$$\phi(x) \leq \psi(x)$$

for later times as long as both functions remain defined.
The basic idea is that, if the two graphs were to cross for the first time, their derivatives at the first crossing point would contradict the assumed differential inequality.

2.19 A Scalar Comparison Theorem

Theorem 2.19.1 (Comparison Theorem)**.** Suppose $f(x, y)$ is continuous and locally Lipschitz with respect to y. Let ϕ and ψ satisfy

$$\phi'(x) \leq f(x, \phi(x))$$

and

$$\psi'(x) = f(x, \psi(x))$$

on an interval I. If

$$\phi(x_0) < \psi(x_0),$$

then

$$\phi(x) < \psi(x)$$

for every later point of I for which both functions are defined.

A reversed version holds when all inequalities are reversed.

Remark 2.19.2. Comparison theorems are particularly useful for estimating nonlinear solutions by simpler differential equations whose behavior is already known.

2.20 Grönwall-Type Estimates

An important special comparison principle concerns integral inequalities.

Lemma 2.20.1 (Grönwall Inequality)**.** Suppose $u(t) \geq 0$ is continuous and satisfies

$$u(t) \leq A + B \int_{t_0}^{t} u(s)\, ds$$

for $t \geq t_0$, where $A, B \geq 0$. Then

$$u(t) \leq Ae^{B(t-t_0)}.$$

Proof. Define

$$v(t) = A + B \int_{t_0}^{t} u(s)\, ds.$$

Then $u(t) \leq v(t)$ and

$$v'(t) = Bu(t) \leq Bv(t).$$

Therefore

$$\frac{d}{dt}\left(e^{-Bt}v(t)\right) \leq 0.$$

Hence

$$v(t) \leq v(t_0)e^{B(t-t_0)} = Ae^{B(t-t_0)}.$$

Since $u \leq v$, the result follows. □

2.21 Linear Control and Global Existence

For nonlinear equations, global existence often follows from an estimate showing that the solution cannot grow too quickly.

Theorem 2.21.1 (Linear Control)**.** Consider

$$y' = f(x, y)$$

on a strip

$$D = \{(x, y) : \alpha < x < \beta, \quad y \in \mathbb{R}\}.$$

Suppose f is continuous, locally Lipschitz with respect to y, and there exist nonnegative continuous functions $a(x)$ and $b(x)$ such that

$$|f(x, y)| \leq a(x)\,|y| + b(x)$$

throughout D.
Then every solution through a point of D can be continued throughout the entire interval

$$(\alpha, \beta).$$

Proof. Let y be a solution and restrict attention first to a compact interval

$$[x_0, T] \subset (\alpha, \beta).$$

There exist constants $A, B > 0$ such that

$$a(x) \le A, \qquad b(x) \le B$$

on this interval.
From

$$y(x) = y_0 + \int_{x_0}^{x} f(s, y(s))\, ds,$$

we obtain

$$|y(x)| \le |y_0| + \int_{x_0}^{x} (A\,|y(s)| + B)\, ds.$$

Thus

$$|y(x)| \le |y_0| + B(T - x_0) + A \int_{x_0}^{x} |y(s)|\ ds.$$

By Grönwall's inequality,

$$|y(x)| \le [|y_0| + B(T - x_0)]\, e^{A(T - x_0)}.$$

Hence the solution remains bounded on every compact subinterval of (α, β). It therefore cannot terminate at an interior point of the strip, and the extension theorem gives continuation throughout the entire interval. □

Remark 2.21.2. The key distinction between nonlinear and linear growth is visible here. A bound of the form

$$|f(x, y)| \le a(x)\,|y| + b(x)$$

prevents finite-time blow-up. Growth substantially faster than linear in y, such as $f(y) = y^2$, may permit finite-time blow-up.

2.22 Summary of the First-Order Theory

The theory developed in this chapter separates several logically different questions.
First, the phase-line method studies the qualitative behavior of autonomous equations without requiring explicit solutions.
Second, the existence and uniqueness theorem answers a local question: under continuity and a Lipschitz condition, a unique solution exists near the initial point.
Third, the extension theorem determines how a local solution develops into a maximal solution. A solution can fail to continue only by approaching the boundary of the domain or by becoming unbounded.
Finally, comparison and linear-control estimates provide tools for proving global existence and bounding solutions.
These ideas will reappear repeatedly. For higher-order equations and systems, existence and uniqueness are formulated in a larger phase space, while the qualitative viewpoint eventually develops into the study of stability, phase portraits, and nonlinear dynamical systems.

Chapter 3

Linear Differential Equations of Higher Order

3.1 Second-Order Linear Differential Equations

A second-order linear differential equation has the form

$$a_0(t)y'' + a_1(t)y' + a_2(t)y = g(t).$$

Whenever $a_0(t) \neq 0$, we divide by $a_0(t)$ and write it in standard form as

$$y'' + p(t)y' + q(t)y = g(t).$$

It is convenient to introduce the differential operator

$$L = D^2 + p(t)D + q(t)I,$$

where D denotes differentiation and I is the identity operator. Then the equation becomes

$$L(y) = g(t).$$

If $g(t) = 0$, the equation is called *homogeneous*:

$$L(y) = 0.$$

If $g(t) \neq 0$, it is called *nonhomogeneous*.

Theorem 3.1.1 (Existence and Uniqueness)**.** Let p, q, g be continuous on an open interval I, and let $t_0 \in I$. Then the initial value problem

$$\begin{cases} y'' + p(t)y' + q(t)y = g(t), \\ y(t_0) = y_0, \\ y'(t_0) = y_1 \end{cases}$$

has a unique solution defined throughout I.

Remark 3.1.2. For a linear equation, the interval of existence is determined by the interval on which the coefficients are continuous. This is stronger than the typical local result for nonlinear equations.

Example 3.1.3. Consider

$$(t^2 - 3t)y'' + ty' - (t + 3)y = 0,$$

with initial conditions prescribed at $t = 1$.
After division by $t^2 - 3t = t(t - 3)$, the coefficients are continuous away from $t = 0$ and $t = 3$. Since the initial point is $t = 1$, the largest interval containing 1 on which the standard-form coefficients are continuous is

$$(0, 3).$$

Hence the unique solution exists throughout $(0, 3)$.

3.2 Superposition Principle

The homogeneous equation

$$L(y) = 0$$

has an important linear structure.

Theorem 3.2.1 (Principle of Superposition)**.** If y_1 and y_2 are solutions of

$$L(y) = 0,$$

then for any constants $c_1, c_2 \in \mathbb{R}$,

$$c_1 y_1 + c_2 y_2$$

is also a solution.

Proof. Since L is linear,

$$L(c_1 y_1 + c_2 y_2) = c_1 L(y_1) + c_2 L(y_2) = 0.$$

□

Thus the set of solutions of a second-order homogeneous linear equation forms a vector space.

3.3 Wronskian and Linear Independence

Definition 3.3.1. For $y_1, y_2 \in C^1(I)$, the *Wronskian* is defined by

$$W[y_1, y_2](t) = \begin{vmatrix} y_1(t) & y_2(t) \\ y_1'(t) & y_2'(t) \end{vmatrix}.$$

Thus

$$W[y_1, y_2] = y_1 y_2' - y_1' y_2.$$

Definition 3.3.2. The functions y_1 and y_2 are linearly dependent on I if there exist constants c_1, c_2, not both zero, such that

$$c_1 y_1(t) + c_2 y_2(t) = 0$$

for every $t \in I$. Otherwise they are linearly independent.

Theorem 3.3.3. Let y_1, y_2 be two solutions of

$$y'' + p(t)y' + q(t)y = 0$$

on I. Then either

$$W[y_1, y_2](t) = 0$$

for every $t \in I$, or

$$W[y_1, y_2](t) \neq 0$$

for every $t \in I$.

Proof. Let

$$W = y_1 y_2' - y_1' y_2.$$

Then

$$W' = y_1 y_2'' - y_1'' y_2.$$

Since

$$y_i'' = -p(t) y_i' - q(t) y_i,$$

we obtain

$$\begin{aligned} W' &= y_1(-p y_2' - q y_2) - (-p y_1' - q y_1) y_2 \\ &= -p(y_1 y_2' - y_1' y_2) \\ &= -pW. \end{aligned}$$

Thus

$$W' + p(t) W = 0,$$

so

$$W(t) = C e^{-\int p(t)\, dt}.$$

Hence either $C = 0$, in which case $W \equiv 0$, or $C \neq 0$, in which case W never vanishes. □

This formula is usually called *Abel's formula*.

3.4 Wronskian Criterion

Theorem 3.4.1. Let y_1, y_2 be solutions of a second-order homogeneous linear equation on I. Then the following statements are equivalent:

1. y_1, y_2 are linearly independent;
2. $W[y_1, y_2](t_0) \neq 0$ for some $t_0 \in I$;
3. $W[y_1, y_2](t) \neq 0$ for every $t \in I$.

Proof. Suppose

$$W[y_1, y_2](t_0) = 0.$$

Then the linear system

$$\begin{cases} c_1 y_1(t_0) + c_2 y_2(t_0) = 0, \\ c_1 y_1'(t_0) + c_2 y_2'(t_0) = 0 \end{cases}$$

has a nonzero solution (c_1, c_2).
Define

$$z(t) = c_1 y_1(t) + c_2 y_2(t).$$

Then z solves the homogeneous equation and satisfies

$$z(t_0) = 0, \qquad z'(t_0) = 0.$$

By uniqueness,

$$z(t) \equiv 0.$$

Thus y_1, y_2 are linearly dependent.
The converse follows immediately from the definition of linear dependence. □

3.5 Fundamental Solutions and General Solution

Definition 3.5.1. Two linearly independent solutions y_1, y_2 of

$$L(y) = 0$$

form a *fundamental set of solutions*.

Theorem 3.5.2 (General Solution)**.** If y_1, y_2 form a fundamental set of solutions of

$$L(y) = 0,$$

then every solution has the form

$$y(t) = c_1 y_1(t) + c_2 y_2(t).$$

Proof. Let y be any solution. Choose c_1, c_2 such that

$$\begin{cases} c_1 y_1(t_0) + c_2 y_2(t_0) = y(t_0), \\ c_1 y_1'(t_0) + c_2 y_2'(t_0) = y'(t_0). \end{cases}$$

Since

$$W[y_1, y_2](t_0) \neq 0,$$

this system has a unique solution.
Now both y and

$$c_1 y_1 + c_2 y_2$$

solve the same differential equation with the same initial values. By the existence and uniqueness theorem, they are identical. □

Thus the solution space of a second-order homogeneous linear equation is two-dimensional.

3.6 Constant-Coefficient Equations

Consider

$$ay'' + by' + cy = 0, \qquad a \neq 0.$$

We look for an exponential solution

$$y = e^{rt}.$$

Then

$$y' = re^{rt}, \qquad y'' = r^2 e^{rt}.$$

Substitution gives

$$(ar^2 + br + c)e^{rt} = 0.$$

Since $e^{rt} \neq 0$, we obtain the *characteristic equation*

$$ar^2 + br + c = 0.$$

The structure of the solution depends on the roots of this polynomial.

3.7 Distinct Real Characteristic Roots

Suppose the characteristic equation has two distinct real roots $r_1 \neq r_2$. Then

$$y_1 = e^{r_1 t}, \qquad y_2 = e^{r_2 t}$$

are solutions.
Their Wronskian is

$$W[y_1, y_2] = (r_2 - r_1)e^{(r_1+r_2)t},$$

which is nonzero. Therefore they are linearly independent.
Hence the general solution is

$$y(t) = c_1 e^{r_1 t} + c_2 e^{r_2 t}.$$

3.8 Complex Characteristic Roots

Suppose

$$r_{1,2} = \alpha \pm i\beta, \qquad \beta \neq 0.$$

The corresponding complex solutions are

$$e^{(\alpha+i\beta)t}, \qquad e^{(\alpha-i\beta)t}.$$

By Euler's formula,

$$e^{(\alpha+i\beta)t} = e^{\alpha t}\Big(\cos\beta t + i\sin\beta t\Big).$$

Since the differential equation has real coefficients, the real and imaginary parts are real-valued solutions:

$$y_1(t) = e^{\alpha t}\cos\beta t,$$

and

$$y_2(t) = e^{\alpha t}\sin\beta t.$$

Thus

$$y(t) = e^{\alpha t}\left(c_1\cos\beta t + c_2\sin\beta t\right).$$

Remark 3.8.1. The real part α controls exponential growth or decay, while the imaginary part β determines the oscillation frequency. This interpretation will reappear in the study of linear systems and phase portraits.

3.9 Repeated Characteristic Roots

Suppose the characteristic equation has a repeated root

$$r_1 = r_2 = r.$$

One solution is

$$y_1 = e^{rt}.$$

A second independent solution is

$$y_2 = te^{rt}.$$

Hence

$$y(t) = (c_1 + c_2 t)e^{rt}.$$

This phenomenon is the first appearance of a general principle: repeated algebraic roots produce polynomial factors in t.

3.10 Reduction of Order

The repeated-root formula suggests a more general method. Suppose one nontrivial solution y_1 of

$$y'' + p(t)y' + q(t)y = 0$$

is already known. We look for another solution of the form

$$y_2 = v(t)y_1(t).$$

Then

$$y_2' = v'y_1 + vy_1',$$

and

$$y_2'' = v''y_1 + 2v'y_1' + vy_1''.$$

Substitution gives

$$v''y_1 + v'\Big(2y_1' + py_1\Big) + v\Big(y_1'' + py_1' + qy_1\Big) = 0.$$

Since y_1 is a solution,

$$y_1'' + py_1' + qy_1 = 0.$$

Therefore

$$v''y_1 + v'\Big(2y_1' + py_1\Big) = 0.$$

Let

$$w = v'.$$

Then

$$w'y_1 + w\Big(2y_1' + py_1\Big) = 0,$$

which is a first-order linear equation for w.
Dividing by y_1,

$$w' + \left(2\frac{y_1'}{y_1} + p\right)w = 0.$$

Hence

$$w = C\frac{e^{-\int p(t)\,dt}}{y_1^2(t)}.$$

Since $w = v'$,

$$v(t) = C\int \frac{e^{-\int p(t)\,dt}}{y_1^2(t)}\,dt.$$

Thus a second solution may be taken as

$$y_2(t) = y_1(t)\int \frac{e^{-\int p(t)\,dt}}{y_1^2(t)}\,dt.$$

Remark 3.10.1. Reduction of order converts a second-order homogeneous equation into a first-order equation once one nontrivial solution is known.

3.11 Nonhomogeneous Linear Equations

Consider

$$L(y) = g(t).$$

Let y_h denote the general solution of the homogeneous equation

$$L(y) = 0,$$

and let y_p be one particular solution of

$$L(y) = g(t).$$

Theorem 3.11.1. The general solution of the nonhomogeneous equation is

$$y = y_h + y_p.$$

Proof. If y_h solves $L(y) = 0$ and y_p solves $L(y) = g$, then

$$L(y_h + y_p) = L(y_h) + L(y_p) = g.$$

Conversely, if y is any solution of $L(y) = g$, then

$$L(y - y_p) = L(y) - L(y_p) = 0.$$

Thus $y - y_p$ is a homogeneous solution. □

3.12 Method of Undetermined Coefficients

For constant-coefficient equations

$$ay'' + by' + cy = g(t),$$

a particular solution can often be guessed when $g(t)$ is composed of polynomials, exponentials, sines, cosines, or their products.
The principle is to choose a trial function containing all terms that remain within the same finite-dimensional family under differentiation.
For example:

$$g(t) = P_m(t)$$

suggests

$$y_p = A_m t^m + \cdots + A_1 t + A_0.$$

If

$$g(t) = e^{\lambda t},$$

we try

$$y_p = Ae^{\lambda t}.$$

If

$$g(t) = A\cos\omega t + B\sin\omega t,$$

we try

$$y_p = C\cos\omega t + D\sin\omega t.$$

For

$$g(t) = e^{\lambda t}\left(P_m(t)\cos\omega t + Q_m(t)\sin\omega t\right),$$

the trial solution has the same general form.

3.13 The Resonance Modification

A trial function may already belong to the homogeneous solution space. In that case, direct substitution cannot determine a particular solution.
Suppose the natural trial function corresponds to a characteristic root r_0 of multiplicity s. Then multiply the trial function by

$$t^s.$$

For example, if

$$y'' - 2y' + y = e^t,$$

then the characteristic polynomial is

$$(r-1)^2.$$

Since e^t corresponds to the repeated root $r = 1$, the naive trial Ae^t belongs to the homogeneous solution space. Therefore the correct trial is

$$y_p = At^2e^t.$$

Remark 3.13.1. The power of t is determined by the multiplicity with which the forcing term overlaps the homogeneous solution.

3.14 Variation of Parameters for Second-Order Equations

The method of undetermined coefficients works only for special forms of $g(t)$. Variation of parameters is more general.
Consider

$$y'' + p(t)y' + q(t)y = g(t),$$

and suppose y_1, y_2 form a fundamental set of solutions of the homogeneous equation.
We seek a particular solution of the form

$$y_p = u_1(t)y_1(t) + u_2(t)y_2(t).$$

Impose the auxiliary condition

$$u_1'y_1 + u_2'y_2 = 0.$$

Then

$$y_p' = u_1y_1' + u_2y_2',$$

and

$$y_p'' = u_1'y_1' + u_2'y_2' + u_1y_1'' + u_2y_2''.$$

Substitution into the differential equation gives

$$u_1'y_1' + u_2'y_2' = g(t).$$

Therefore

$$\begin{cases} u_1'y_1 + u_2'y_2 = 0, \\ u_1'y_1' + u_2'y_2' = g(t). \end{cases}$$

By Cramer's rule,

$$u_1' = -\frac{y_2g}{W}, \qquad u_2' = \frac{y_1g}{W},$$

where

$$W = W[y_1, y_2].$$

Thus

$$u_1 = -\int \frac{y_2(t)g(t)}{W(t)}\,dt,$$

and

$$u_2 = \int \frac{y_1(t)g(t)}{W(t)}\,dt.$$

Hence one particular solution is

$$y_p = -y_1(t)\int \frac{y_2(t)g(t)}{W(t)}\,dt + y_2(t)\int \frac{y_1(t)g(t)}{W(t)}\,dt.$$

Remark 3.14.1. Variation of parameters has the same conceptual origin as the first-order method from Chapter 1: constants in the homogeneous solution are replaced by functions.

3.15 Mechanical Vibrations

A standard application of second-order linear equations is the mass–spring system.
Let

$$y(t)$$

denote the displacement from equilibrium. Newton's second law gives

$$my'' = F_{\text{spring}} + F_{\text{damping}} + F_{\text{external}}.$$

For a linear spring,

$$F_{\text{spring}} = -ky,$$

where $k > 0$ is the spring constant.
For viscous damping,

$$F_{\text{damping}} = -cy',$$

where $c \geq 0$.
If the external force is $F(t)$, then

$$my'' + cy' + ky = F(t).$$

3.16 Free Undamped Motion

If $c = 0$ and $F(t) = 0$, then

$$my'' + ky = 0.$$

Dividing by m,

$$y'' + \omega_0^2 y = 0,$$

where

$$\omega_0 = \sqrt{\frac{k}{m}}$$

is the natural angular frequency.
The general solution is

$$y(t) = c_1 \cos \omega_0 t + c_2 \sin \omega_0 t.$$

Equivalently,

$$y(t) = R\cos(\omega_0 t - \delta),$$

where R is the amplitude and δ is the phase shift.
The motion is periodic with period

$$T = \frac{2\pi}{\omega_0}.$$

3.17 Free Damped Motion

Now consider

$$my'' + cy' + ky = 0.$$

The characteristic equation is

$$mr^2 + cr + k = 0,$$

with roots

$$r = \frac{-c \pm \sqrt{c^2 - 4mk}}{2m}.$$

The behavior depends on the discriminant

$$c^2 - 4mk.$$

If

$$c^2 > 4mk,$$

the system is *overdamped*. There are two distinct negative real roots, and the displacement approaches equilibrium without oscillation.
If

$$c^2 = 4mk,$$

the system is *critically damped*. There is one repeated negative root, and

$$y(t) = (c_1 + c_2 t)e^{-ct/(2m)}.$$

If

$$c^2 < 4mk,$$

the system is *underdamped*. Write

$$\alpha = -\frac{c}{2m},$$

and

$$\omega = \frac{\sqrt{4mk - c^2}}{2m}.$$

Then

$$y(t) = e^{\alpha t}\left(c_1 \cos \omega t + c_2 \sin \omega t\right).$$

Thus the system oscillates with exponentially decaying amplitude.

3.18 Forced Oscillations

Consider the undamped forced equation

$$y'' + \omega_0^2 y = A\cos\omega t.$$

The behavior depends on whether the forcing frequency ω coincides with the natural frequency ω_0.
Suppose first that

$$\omega \neq \omega_0.$$

Try

$$y_p = C\cos\omega t.$$

Then

$$y_p'' = -C\omega^2\cos\omega t.$$

Hence

$$C\left(\omega_0^2 - \omega^2\right)\cos\omega t = A\cos\omega t.$$

Therefore

$$C = \frac{A}{\omega_0^2 - \omega^2}.$$

Thus

$$y_p = \frac{A}{\omega_0^2 - \omega^2}\cos\omega t.$$

3.19 Resonance

If

$$\omega = \omega_0,$$

the forcing term is itself a homogeneous solution. Hence the previous trial fails.
We instead try

$$y_p = Ct\sin\omega_0 t.$$

Substitution gives

$$C = \frac{A}{2\omega_0}.$$

Therefore

$$y_p = \frac{A}{2\omega_0}t\sin\omega_0 t.$$

The factor t causes the amplitude to grow without bound. This phenomenon is called *resonance*.

Remark 3.19.1. Resonance is another manifestation of the same algebraic phenomenon seen in the method of undetermined coefficients: when the forcing term belongs to the homogeneous solution space, multiplication by t is required.

3.20 General n-th Order Linear Equations

We now consider the general equation

$$y^{(n)} + p_1(t)y^{(n-1)} + \cdots + p_{n-1}(t)y' + p_n(t)y = g(t).$$

Define

$$L(y) = y^{(n)} + p_1(t)y^{(n-1)} + \cdots + p_n(t)y.$$

Theorem 3.20.1 (Existence and Uniqueness)**.** Suppose

$$p_1, \ldots, p_n, g$$

are continuous on an interval I, and $t_0 \in I$. Then for any initial values

$$y(t_0) = a_0, \quad y'(t_0) = a_1, \quad \ldots, \quad y^{(n-1)}(t_0) = a_{n-1},$$

the problem

$$L(y) = g$$

has a unique solution defined throughout I.

The equation can be rewritten as a system by setting

$$x_1 = y, \qquad x_2 = y', \qquad \ldots, \qquad x_n = y^{(n-1)}.$$

Then

$$x_1' = x_2,$$
$$x_2' = x_3,$$

and so on, while

$$x_n' = g(t) - p_1(t)x_n - \cdots - p_n(t)x_1.$$

Thus the theory of higher-order equations is naturally connected with systems of first-order equations.

3.21 Higher-Order Wronskian

For functions

$$y_1, \ldots, y_n,$$

define

$$W[y_1, \ldots, y_n](t) = \det \begin{pmatrix} y_1 & y_2 & \cdots & y_n \\ y_1' & y_2' & \cdots & y_n' \\ \vdots & \vdots & \ddots & \vdots \\ y_1^{(n-1)} & y_2^{(n-1)} & \cdots & y_n^{(n-1)} \end{pmatrix}.$$

For solutions of an n-th order homogeneous linear equation, the following conditions are equivalent:

1. $y_1, \ldots, y_n$ are linearly independent;
2. $W[y_1, \ldots, y_n](t_0) \neq 0$ for some $t_0 \in I$;
3. $W[y_1, \ldots, y_n](t) \neq 0$ for every $t \in I$;
4. every solution can be written uniquely as

$$y = c_1 y_1 + \cdots + c_n y_n.$$

Hence the solution space of an n-th order homogeneous linear equation is n-dimensional.

3.22 Constant-Coefficient Equations of Higher Order

Consider

$$a_0 y^{(n)} + a_1 y^{(n-1)} + \cdots + a_{n-1} y' + a_n y = 0.$$

Substituting

$$y = e^{rt}$$

gives the characteristic polynomial

$$P(r) = a_0 r^n + a_1 r^{n-1} + \cdots + a_{n-1} r + a_n.$$

Suppose $r \in \mathbb{R}$ is a root of multiplicity m. Then the corresponding solutions are

$$e^{rt}, \quad te^{rt}, \quad \ldots, \quad t^{m-1} e^{rt}.$$

Suppose

$$r = \alpha + i\beta, \qquad \beta \neq 0,$$

is a complex root of multiplicity m. Since the coefficients are real,

$$\alpha - i\beta$$

is also a root of multiplicity m. The corresponding real solutions are

$$e^{\alpha t} \cos \beta t, \qquad e^{\alpha t} \sin \beta t,$$

$$te^{\alpha t} \cos \beta t, \qquad te^{\alpha t} \sin \beta t,$$

and, in general,

$$t^k e^{\alpha t} \cos \beta t, \qquad t^k e^{\alpha t} \sin \beta t,$$

for

$$k = 0, \ldots, m-1.$$

3.23 Variation of Parameters in the General Case

Suppose

$$y_1, \ldots, y_n$$

form a fundamental set of solutions of

$$L(y) = 0.$$

The homogeneous solution is

$$y_h = c_1 y_1 + \cdots + c_n y_n.$$

To find a particular solution of

$$L(y) = g(t),$$

replace the constants by functions:

$$y_p = u_1(t) y_1(t) + \cdots + u_n(t) y_n(t).$$

Impose

$$\sum_{j=1}^{n} u_j'(t) y_j^{(k)}(t) = 0, \qquad k = 0, \ldots, n-2,$$

together with

$$\sum_{j=1}^{n} u_j'(t) y_j^{(n-1)}(t) = g(t).$$

This gives the matrix system

$$\begin{pmatrix} y_1 & \cdots & y_n \\ y_1' & \cdots & y_n' \\ \vdots & \ddots & \vdots \\ y_1^{(n-1)} & \cdots & y_n^{(n-1)} \end{pmatrix} \begin{pmatrix} u_1' \\ u_2' \\ \vdots \\ u_n' \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ g(t) \end{pmatrix}.$$

Since the coefficient matrix has determinant

$$W[y_1, \ldots, y_n](t) \neq 0,$$

the functions $u_1', \ldots, u_n'$ are uniquely determined.
By Cramer's rule,

$$u_j'(t) = \frac{\det W_j(t)}{W(t)},$$

where $W_j(t)$ is obtained from the Wronskian matrix by replacing its j-th column with

$$(0, \ldots, 0, g(t))^T.$$

After integration,

$$u_j(t) = \int u_j'(t)\, dt,$$

and

$$y_p = \sum_{j=1}^{n} u_j(t) y_j(t).$$

Remark 3.23.1. The higher-order theory has now become visibly linear algebraic. A fundamental set of scalar solutions plays the same role that a basis plays in a vector space, while the Wronskian tests whether that basis is independent. In the next chapter, this structure is rewritten directly in vector and matrix form.

Chapter 4

Systems of First-Order Linear Differential Equations

4.1 Vector-Valued and Matrix-Valued Functions

A system of first-order differential equations can be written in the form

$$\begin{cases} x_1' = f_1(t, x_1, \dots, x_n), \\ x_2' = f_2(t, x_1, \dots, x_n), \\ \vdots \\ x_n' = f_n(t, x_1, \dots, x_n). \end{cases}$$

Introducing the vector

$$\boldsymbol{x}(t) = \begin{pmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_n(t) \end{pmatrix},$$

and

$$\boldsymbol{f}(t, \boldsymbol{x}) = \begin{pmatrix} f_1(t, \boldsymbol{x}) \\ f_2(t, \boldsymbol{x}) \\ \vdots \\ f_n(t, \boldsymbol{x}) \end{pmatrix},$$

the system can be written compactly as

$$\boldsymbol{x}'(t) = \boldsymbol{f}(t, \boldsymbol{x}(t)).$$

A matrix-valued function is a function

$$A : I \longrightarrow M_{m\times n}(\mathbb{R})$$

of the form

$$A(t) = \Big(a_{ij}(t)\Big).$$

It is continuous or differentiable if every entry $a_{ij}(t)$ is continuous or differentiable.
If $A(t)$ and $B(t)$ are differentiable matrix-valued functions, then

$$(A + B)' = A' + B',$$

and

$$(AB)' = A'B + AB'.$$

If $\boldsymbol{x}(t)$ is a differentiable vector-valued function, then

$$\frac{d}{dt}\Big(A(t)\boldsymbol{x}(t)\Big) = A'(t)\boldsymbol{x}(t) + A(t)\boldsymbol{x}'(t).$$

These rules are formally identical to the usual product rule, provided the order of matrix multiplication is preserved.

4.2 Existence and Uniqueness for Systems

Consider the initial value problem

$$\begin{cases} \boldsymbol{x}' = \boldsymbol{f}(t, \boldsymbol{x}), \\ \boldsymbol{x}(t_0) = \boldsymbol{x}_0. \end{cases}$$

Theorem 4.2.1 (Existence and Uniqueness)**.** Suppose $\boldsymbol{f}(t, \boldsymbol{x})$ is continuous in a rectangle

$$R = \{(t, \boldsymbol{x}) : |t - t_0| \leq a, \quad \|\boldsymbol{x} - \boldsymbol{x}_0\| \leq b\},$$

and suppose there exists $L > 0$ such that

$$\|\boldsymbol{f}(t, \boldsymbol{x}_1) - \boldsymbol{f}(t, \boldsymbol{x}_2)\| \leq L \|\boldsymbol{x}_1 - \boldsymbol{x}_2\|$$

for all relevant points in R.
Then the initial value problem has a unique local solution.

Thus the existence and uniqueness theory of Chapter 2 extends naturally from scalar equations to vector-valued equations.

4.3 Linear Systems

A first-order linear system has the form

$$\boldsymbol{x}'(t) = P(t)\boldsymbol{x}(t) + \boldsymbol{g}(t),$$

where $P(t)$ is an $n \times n$ matrix-valued function and $\boldsymbol{g}(t)$ is an n-dimensional vector-valued function.
The corresponding homogeneous system is

$$\boldsymbol{x}'(t) = P(t)\boldsymbol{x}(t).$$

Theorem 4.3.1. Suppose $P(t)$ and $\boldsymbol{g}(t)$ are continuous on an interval I. Then for every $t_0 \in I$ and every initial vector $\boldsymbol{x}_0 \in \mathbb{R}^n$, the problem

$$\begin{cases} \boldsymbol{x}' = P(t)\boldsymbol{x} + \boldsymbol{g}(t), \\ \boldsymbol{x}(t_0) = \boldsymbol{x}_0 \end{cases}$$

has a unique solution defined throughout I.

Remark 4.3.2. As in scalar linear equations, continuity of the coefficients gives existence throughout the entire interval of continuity. This is one of the major differences between linear and general nonlinear systems.

4.4 Principle of Superposition

Consider the homogeneous system

$$\boldsymbol{x}' = P(t)\boldsymbol{x}.$$

Theorem 4.4.1 (Principle of Superposition)**.** If

$$\boldsymbol{x}^{(1)}(t), \ldots, \boldsymbol{x}^{(k)}(t)$$

are solutions and $c_1, \ldots, c_k$ are constants, then

$$\boldsymbol{x}(t) = \sum_{j=1}^{k} c_j \boldsymbol{x}^{(j)}(t)$$

is also a solution.

Proof. Since each $\boldsymbol{x}^{(j)}$ satisfies

$$\boldsymbol{x}^{(j)\prime} = P(t)\boldsymbol{x}^{(j)},$$

we have

$$\begin{aligned}\boldsymbol{x}' &= \sum_{j=1}^{k} c_j \boldsymbol{x}^{(j)\prime} \\ &= \sum_{j=1}^{k} c_j P(t)\boldsymbol{x}^{(j)} \\ &= P(t)\sum_{j=1}^{k} c_j \boldsymbol{x}^{(j)} \\ &= P(t)\boldsymbol{x}.\end{aligned}$$

□

Therefore the solution set of an n-dimensional homogeneous linear system forms a vector space.

4.5 Wronskian for Linear Systems

Let

$$\boldsymbol{x}^{(1)}(t), \ldots, \boldsymbol{x}^{(n)}(t)$$

be solutions of

$$\boldsymbol{x}' = P(t)\boldsymbol{x}.$$

Place them as columns of a matrix:

$$X(t) = \begin{pmatrix} | & | & & | \\ \boldsymbol{x}^{(1)}(t) & \boldsymbol{x}^{(2)}(t) & \cdots & \boldsymbol{x}^{(n)}(t) \\ | & | & & | \end{pmatrix}.$$

Definition 4.5.1. The *Wronskian* of $\boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(n)}$ is

$$W(t) = \det X(t).$$

Theorem 4.5.2. For solutions of a homogeneous linear system, the following statements are equivalent:

1. $\boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(n)}$ are linearly independent;
2. there exists $t_0 \in I$ such that $W(t_0) \neq 0$;
3. $W(t) \neq 0$ for every $t \in I$;
4. every solution can be written uniquely as

$$\boldsymbol{x}(t) = c_1\boldsymbol{x}^{(1)}(t) + \cdots + c_n\boldsymbol{x}^{(n)}(t).$$

Proof. Suppose first that

$$W(t_0) = 0.$$

Then the columns

$$\boldsymbol{x}^{(1)}(t_0), \ldots, \boldsymbol{x}^{(n)}(t_0)$$

are linearly dependent. Hence there exist constants $c_1, \dots, c_n$, not all zero, such that

$$\sum_{j=1}^{n} c_j \boldsymbol{x}^{(j)}(t_0) = \mathbf{0}.$$

Define

$$\boldsymbol{z}(t) = \sum_{j=1}^{n} c_j \boldsymbol{x}^{(j)}(t).$$

Then $\boldsymbol{z}$ solves the homogeneous system and satisfies

$$\boldsymbol{z}(t_0) = \mathbf{0}.$$

By uniqueness,

$$\boldsymbol{z}(t) \equiv \mathbf{0}.$$

Thus the solutions are linearly dependent.
Conversely, if the solutions are linearly dependent, then their columns are linearly dependent at every t, so $W(t) = 0$. □

4.6 Abel's Formula for Systems

The scalar Abel identity from Chapter 3 has a direct matrix analogue.

Theorem 4.6.1 (Abel's Formula)**.** Suppose

$$\boldsymbol{x}' = P(t)\boldsymbol{x},$$

where $P(t)$ is continuous on I. If

$$W(t) = W[\boldsymbol{x}^{(1)}, \dots, \boldsymbol{x}^{(n)}](t),$$

then

$$W(t) = W(t_0) \exp\left(\int_{t_0}^{t} \operatorname{tr} P(s)\, ds \right).$$

Therefore the Wronskian is either identically zero or never zero on I.

Remark 4.6.2. For a scalar n-th order equation, Abel's formula involves the coefficient of $y^{(n-1)}$. For a first-order system, that quantity is replaced naturally by the trace of the coefficient matrix.

4.7 Fundamental Sets of Solutions

Definition 4.7.1. A collection

$$\boldsymbol{x}^{(1)}, \dots, \boldsymbol{x}^{(n)}$$

of linearly independent solutions of

$$\boldsymbol{x}' = P(t)\boldsymbol{x}$$

is called a *fundamental set of solutions*.

For every $t_0 \in I$, let $\boldsymbol{x}^{(j)}(t)$ be the solution satisfying

$$\boldsymbol{x}^{(j)}(t_0) = \boldsymbol{e}_j,$$

where

$$\boldsymbol{e}_1, \ldots, \boldsymbol{e}_n$$

are the standard basis vectors of $\mathbb{R}^n$.
Then

$$W(t_0) = \det I = 1,$$

so these solutions form a fundamental set.
Consequently, an n-dimensional linear homogeneous system has an n-dimensional solution space.

4.8 Fundamental Matrices

Definition 4.8.1. If

$$\boldsymbol{x}^{(1)}(t), \ldots, \boldsymbol{x}^{(n)}(t)$$

form a fundamental set of solutions, then

$$\Phi(t) = \begin{pmatrix} | & | & & | \\ \boldsymbol{x}^{(1)}(t) & \boldsymbol{x}^{(2)}(t) & \cdots & \boldsymbol{x}^{(n)}(t) \\ | & | & & | \end{pmatrix}$$

is called a *fundamental matrix*.

Since its columns solve

$$\boldsymbol{x}' = P(t)\boldsymbol{x},$$

we have

$$\Phi'(t) = P(t)\Phi(t).$$

Since the columns are linearly independent,

$$\det \Phi(t) \neq 0,$$

so $\Phi(t)$ is invertible for every $t \in I$.
The general solution is

$$\boldsymbol{x}(t) = \Phi(t)\boldsymbol{c},$$

where $\boldsymbol{c} \in \mathbb{R}^n$ is an arbitrary constant vector.

4.9 Standard Fundamental Matrix

Fix $t_0 \in I$.

Definition 4.9.1. The *standard fundamental matrix* at t_0 is the fundamental matrix satisfying

$$\Phi(t_0) = I.$$

If $X(t)$ is any fundamental matrix, then

$$\Phi(t) = X(t)X(t_0)^{-1}$$

is the standard fundamental matrix at t_0.

Indeed,

$$\Phi(t_0) = X(t_0)X(t_0)^{-1} = I.$$

For the initial value problem

$$\begin{cases} \boldsymbol{x}' = P(t)\boldsymbol{x}, \\ \boldsymbol{x}(t_0) = \boldsymbol{x}_0, \end{cases}$$

the solution is

$$\boldsymbol{x}(t) = X(t)X(t_0)^{-1}\boldsymbol{x}_0.$$

Thus the fundamental matrix plays the role of the scalar homogeneous solution in a first-order linear equation.

4.10 Constant-Coefficient Linear Systems

We now consider

$$\boldsymbol{x}' = A\boldsymbol{x},$$

where A is a constant $n \times n$ real matrix.
Suppose $(\lambda, \boldsymbol{v})$ is an eigenpair of A:

$$A\boldsymbol{v} = \lambda\boldsymbol{v}.$$

Then

$$\boldsymbol{x}(t) = e^{\lambda t}\boldsymbol{v}$$

is a solution.
Indeed,

$$\boldsymbol{x}'(t) = \lambda e^{\lambda t}\boldsymbol{v},$$

while

$$A\boldsymbol{x}(t) = e^{\lambda t}A\boldsymbol{v} = \lambda e^{\lambda t}\boldsymbol{v}.$$

Hence eigenvalues and eigenvectors transform the differential equation into an algebraic problem.

4.11 Distinct Real Eigenvalues

Suppose A has n linearly independent real eigenvectors

$$\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n$$

with corresponding real eigenvalues

$$\lambda_1, \ldots, \lambda_n.$$

Then

$$\boldsymbol{x}^{(j)}(t) = e^{\lambda_j t}\boldsymbol{v}_j, \qquad j = 1, \ldots, n,$$

form a fundamental set of solutions.
Indeed,

$$\det \begin{pmatrix} | & & | \\ e^{\lambda_1 t}\boldsymbol{v}_1 & \cdots & e^{\lambda_n t}\boldsymbol{v}_n \\ | & & | \end{pmatrix} = e^{(\lambda_1 + \cdots + \lambda_n)t} \det \begin{pmatrix} | & & | \\ \boldsymbol{v}_1 & \cdots & \boldsymbol{v}_n \\ | & & | \end{pmatrix},$$

which is nonzero because the eigenvectors are linearly independent.
Therefore

$$\boldsymbol{x}(t) = \sum_{j=1}^{n} c_j e^{\lambda_j t}\boldsymbol{v}_j.$$

4.12 Diagonalization

Suppose A is diagonalizable:

$$A = PDP^{-1},$$

where

$$D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n).$$

Set

$$\boldsymbol{x} = P\boldsymbol{y}.$$

Then

$$P\boldsymbol{y}' = AP\boldsymbol{y} = PDP^{-1}P\boldsymbol{y} = PD\boldsymbol{y}.$$

Therefore

$$\boldsymbol{y}' = D\boldsymbol{y}.$$

The system decouples into

$$y_j' = \lambda_j y_j,$$

so

$$y_j(t) = c_j e^{\lambda_j t}.$$

Thus

$$\boldsymbol{x}(t) = P \begin{pmatrix} c_1 e^{\lambda_1 t} \\ \vdots \\ c_n e^{\lambda_n t} \end{pmatrix}.$$

Remark 4.12.1. Diagonalization reveals why the eigenvector method works: in eigenvector coordinates, the system is simply a collection of independent scalar equations.

4.13 Complex Eigenvalues

Suppose A is real and has a complex eigenvalue

$$\lambda = \alpha + i\beta, \qquad \beta \neq 0,$$

with a complex eigenvector

$$\boldsymbol{v} = \boldsymbol{u} + i\boldsymbol{w}.$$

Then

$$\boldsymbol{z}(t) = e^{(\alpha+i\beta)t}(\boldsymbol{u} + i\boldsymbol{w})$$

is a complex-valued solution.
Using Euler's formula,

$$e^{(\alpha+i\beta)t} = e^{\alpha t}\Big(\cos\beta t + i\sin\beta t\Big),$$

we obtain

$$\begin{aligned} \boldsymbol{z}(t) &= e^{\alpha t}\Big(\cos\beta t + i\sin\beta t\Big)(\boldsymbol{u} + i\boldsymbol{w}) \\ &= e^{\alpha t}\left(\boldsymbol{u}\cos\beta t - \boldsymbol{w}\sin\beta t\right) \\ &\quad + ie^{\alpha t}\left(\boldsymbol{u}\sin\beta t + \boldsymbol{w}\cos\beta t\right). \end{aligned}$$

Since A is real, both the real and imaginary parts are real-valued solutions:

$$\boldsymbol{x}^{(1)}(t) = e^{\alpha t}\left(\boldsymbol{u}\cos\beta t - \boldsymbol{w}\sin\beta t\right),$$

and

$$\boldsymbol{x}^{(2)}(t) = e^{\alpha t}\left(\boldsymbol{u}\sin\beta t + \boldsymbol{w}\cos\beta t\right).$$

Remark 4.13.1. The factor $e^{\alpha t}$ determines growth or decay, while β produces rotation or oscillation. This interpretation will become geometrically important when phase portraits are studied in Chapter 5.

4.14 Repeated Eigenvalues and Generalized Eigenvectors

If a repeated eigenvalue does not provide enough linearly independent eigenvectors, ordinary diagonalization is impossible.

Suppose

$$\lambda$$

is an eigenvalue and

$$\boldsymbol{v}_1, \dots, \boldsymbol{v}_m$$

form a chain of generalized eigenvectors satisfying

$$(A - \lambda I)\boldsymbol{v}_1 = 0,$$

and

$$(A - \lambda I)\boldsymbol{v}_{j+1} = \boldsymbol{v}_j.$$

The corresponding solutions contain polynomial factors in t, just as repeated roots do for scalar constant-coefficient equations.

For example, for a chain of length two,

$$\boldsymbol{x}^{(1)}(t) = e^{\lambda t}\boldsymbol{v}_1,$$

and

$$\boldsymbol{x}^{(2)}(t) = e^{\lambda t}\left(t\boldsymbol{v}_1 + \boldsymbol{v}_2\right).$$

Thus repeated characteristic roots in scalar equations and generalized eigenvectors in systems are two forms of the same algebraic phenomenon.

4.15 Matrix Exponential

The eigenvalue method motivates a more intrinsic construction.

Definition 4.15.1. For a constant square matrix A, the *matrix exponential* is defined by

$$e^A = \sum_{k=0}^{\infty} \frac{A^k}{k!}.$$

More generally,

$$e^{At} = \sum_{k=0}^{\infty} \frac{A^k t^k}{k!}.$$

The series converges, and term-by-term differentiation gives

$$\frac{d}{dt}e^{At} = Ae^{At} = e^{At}A.$$

Also,

$$e^{A\cdot 0} = I.$$

Therefore

$$\Phi(t) = e^{At}$$

is the standard fundamental matrix at $t_0 = 0$ for

$$\boldsymbol{x}' = A\boldsymbol{x}.$$

Consequently,

$$\boldsymbol{x}(t) = e^{At}\boldsymbol{x}_0$$

solves

$$\begin{cases} \boldsymbol{x}' = A\boldsymbol{x}, \\ \boldsymbol{x}(0) = \boldsymbol{x}_0. \end{cases}$$

4.16 Basic Properties of the Matrix Exponential

For a constant matrix A,

$$e^{A(t+s)} = e^{At}e^{As}.$$

In particular,

$$(e^{At})^{-1} = e^{-At}.$$

If A and B commute,

$$AB = BA,$$

then

$$e^{A+B} = e^{A}e^{B}.$$

Remark 4.16.1. The commutativity condition is essential. In general,

$$e^{A+B} \neq e^{A}e^{B}$$

when $AB \neq BA$.

4.17 Computing the Matrix Exponential

Several cases are particularly useful.
If

$$A = \operatorname{diag}(\lambda_1, \dots, \lambda_n),$$

then

$$e^{At} = \operatorname{diag}(e^{\lambda_1 t}, \dots, e^{\lambda_n t}).$$

If A is diagonalizable,

$$A = PDP^{-1},$$

then

$$A^k = PD^kP^{-1},$$

so

$$e^{At} = Pe^{Dt}P^{-1}.$$

If A is not diagonalizable, generalized eigenvectors or an equivalent Jordan-form calculation produce polynomial factors multiplying $e^{\lambda t}$.

4.18 A Two-Dimensional Example

Consider

$$\boldsymbol{x}' = \begin{pmatrix} 3 & 1 \\ 0 & -1 \end{pmatrix} \boldsymbol{x}.$$

The eigenvalues are

$$\lambda_1 = 3, \qquad \lambda_2 = -1.$$

For $\lambda_1 = 3$, an eigenvector is

$$\boldsymbol{v}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

For $\lambda_2 = -1$, solve

$$(A + I)\boldsymbol{v} = 0,$$

which gives an eigenvector proportional to

$$\boldsymbol{v}_2 = \begin{pmatrix} -1 \\ 4 \end{pmatrix}.$$

Thus a fundamental set is

$$\boldsymbol{x}^{(1)}(t) = e^{3t} \begin{pmatrix} 1 \\ 0 \end{pmatrix},$$

and

$$\boldsymbol{x}^{(2)}(t) = e^{-t} \begin{pmatrix} -1 \\ 4 \end{pmatrix}.$$

Hence

$$\boldsymbol{x}(t) = c_1 e^{3t} \begin{pmatrix} 1 \\ 0 \end{pmatrix} + c_2 e^{-t} \begin{pmatrix} -1 \\ 4 \end{pmatrix}.$$

One mode grows exponentially while the other decays exponentially. The geometric meaning of this behavior will be discussed through phase portraits in the next chapter.

4.19 Nonhomogeneous Linear Systems

Consider

$$\boldsymbol{x}' = P(t)\boldsymbol{x} + \boldsymbol{g}(t).$$

Let $\Phi(t)$ be a fundamental matrix for the corresponding homogeneous system

$$\boldsymbol{x}' = P(t)\boldsymbol{x}.$$

As in the scalar theory, the general solution consists of a homogeneous part and one particular solution:

$$\boldsymbol{x} = \boldsymbol{x}_h + \boldsymbol{x}_p.$$

4.20 Variation of Parameters for Systems

We seek a particular solution in the form

$$\boldsymbol{x}_p(t) = \Phi(t)\boldsymbol{u}(t),$$

where $\boldsymbol{u}(t)$ is an unknown vector-valued function.
Differentiating,

$$\boldsymbol{x}_p' = \Phi'\boldsymbol{u} + \Phi\boldsymbol{u}'.$$

Since

$$\Phi' = P(t)\Phi,$$

we have

$$\boldsymbol{x}_p' = P(t)\Phi\boldsymbol{u} + \Phi\boldsymbol{u}'.$$

On the other hand,

$$\boldsymbol{x}_p' = P(t)\boldsymbol{x}_p + \boldsymbol{g}(t) = P(t)\Phi\boldsymbol{u} + \boldsymbol{g}(t).$$

Therefore

$$\Phi(t)\boldsymbol{u}'(t) = \boldsymbol{g}(t),$$

and hence

$$\boldsymbol{u}'(t) = \Phi(t)^{-1}\boldsymbol{g}(t).$$

Thus

$$\boldsymbol{u}(t) = \int \Phi(t)^{-1}\boldsymbol{g}(t)\,dt,$$

and a particular solution is

$$\boldsymbol{x}_p(t) = \Phi(t)\int \Phi(t)^{-1}\boldsymbol{g}(t)\,dt.$$

Therefore the general solution is

$$\boxed{\boldsymbol{x}(t) = \Phi(t)\left[\boldsymbol{c} + \int \Phi(t)^{-1}\boldsymbol{g}(t)\,dt\right].}$$

4.21 Variation of Constants Formula

For the initial value problem

$$\begin{cases}\boldsymbol{x}' = P(t)\boldsymbol{x} + \boldsymbol{g}(t),\\ \boldsymbol{x}(t_0) = \boldsymbol{x}_0,\end{cases}$$

let $\Phi(t)$ be any fundamental matrix. Then

$$\boldsymbol{x}(t) = \Phi(t)\Phi(t_0)^{-1}\boldsymbol{x}_0 + \Phi(t)\int_{t_0}^{t} \Phi(s)^{-1}\boldsymbol{g}(s)\,ds.$$

If Φ is the standard fundamental matrix at t_0, so that

$$\Phi(t_0) = I,$$

this reduces to

$$\boldsymbol{x}(t) = \Phi(t)\boldsymbol{x}_0 + \Phi(t)\int_{t_0}^{t} \Phi(s)^{-1}\boldsymbol{g}(s)\,ds.$$

4.22 Constant-Coefficient Nonhomogeneous Systems

For

$$\boldsymbol{x}' = A\boldsymbol{x} + \boldsymbol{g}(t),$$

we may take

$$\Phi(t) = e^{At}.$$

Therefore the solution satisfying

$$\boldsymbol{x}(t_0) = \boldsymbol{x}_0$$

is

$$\boldsymbol{x}(t) = e^{A(t-t_0)}\boldsymbol{x}_0 + \int_{t_0}^{t} e^{A(t-s)}\boldsymbol{g}(s)\,ds.$$

Indeed,

$$e^{At}e^{-As} = e^{A(t-s)},$$

so the variation-of-parameters formula simplifies naturally to this convolution-type expression.

Remark 4.22.1. This formula is the system analogue of the integrating-factor formula for a scalar first-order linear equation. The matrix exponential e^{At} plays exactly the role that the scalar exponential integrating factor plays in one dimension.

4.23 Undetermined Coefficients for Systems

For constant-coefficient systems

$$\boldsymbol{x}' = A\boldsymbol{x} + \boldsymbol{g}(t),$$

a particular solution can sometimes also be found by the method of undetermined coefficients.

If $\boldsymbol{g}(t)$ is polynomial, exponential, trigonometric, or a combination of these, one proposes a vector-valued trial function of the corresponding form and determines its coefficients by substitution.

For example, if

$$\boldsymbol{g}(t) = e^{\lambda t}\boldsymbol{b},$$

one may try

$$\boldsymbol{x}_p(t) = e^{\lambda t}\boldsymbol{v}.$$

Substitution gives

$$\lambda e^{\lambda t}\boldsymbol{v} = Ae^{\lambda t}\boldsymbol{v} + e^{\lambda t}\boldsymbol{b},$$

so

$$(\lambda I - A)\boldsymbol{v} = \boldsymbol{b}.$$

If $\lambda I - A$ is invertible, then

$$\boldsymbol{v} = (\lambda I - A)^{-1}\boldsymbol{b}.$$

If λ is an eigenvalue of A, the trial form must be modified, analogous to resonance in scalar equations.

Remark 4.23.1. Variation of parameters is the general method, whereas undetermined coefficients is usually more efficient when the forcing term has a simple algebraic form.

4.24 From Linear Systems to Phase Portraits

The formula

$$\boldsymbol{x}(t) = e^{At}\boldsymbol{x}_0$$

contains the complete solution of a constant-coefficient homogeneous system. However, for a two-dimensional system, another question is often more informative than the explicit formula:

What do all trajectories look like in the phase plane?

The answer depends mainly on the eigenvalues of A. Real eigenvalues lead to node or saddle behavior, while complex eigenvalues lead to spirals or centers. Moreover, these linear phase portraits provide the local models for nonlinear autonomous systems near equilibrium points.

This geometric viewpoint is the subject of the next chapter.

Chapter 5

Phase Portraits and Nonlinear Autonomous Systems

5.1 Phase Portraits of Planar Linear Systems

Consider the two-dimensional linear autonomous system

$$\boldsymbol{x}' = A\boldsymbol{x}, \qquad \boldsymbol{x} = \begin{pmatrix} x \\ y \end{pmatrix},$$

where A is a real 2×2 matrix.
The solution through an initial point $\boldsymbol{x}_0$ is

$$\boldsymbol{x}(t) = e^{At}\boldsymbol{x}_0.$$

Instead of studying one solution at a time, we now study all solution curves simultaneously in the xy-plane.

Definition 5.1.1. The collection of all solution curves of a planar autonomous system is called its *phase portrait*. Each solution curve is called a *trajectory* or *orbit*.

For the linear system $\boldsymbol{x}' = A\boldsymbol{x}$, the origin is always an equilibrium point since

$$A\mathbf{0} = \mathbf{0}.$$

The geometry near the origin is determined mainly by the eigenvalues of A.

5.2 Two Distinct Real Eigenvalues

Suppose A has two distinct real eigenvalues

$$\lambda_1, \qquad \lambda_2,$$

with linearly independent eigenvectors

$$\boldsymbol{v}_1, \qquad \boldsymbol{v}_2.$$

Then

$$\boldsymbol{x}(t) = c_1 e^{\lambda_1 t}\boldsymbol{v}_1 + c_2 e^{\lambda_2 t}\boldsymbol{v}_2.$$

The eigendirections themselves are invariant trajectories. Indeed, if $\boldsymbol{x}_0 = c\boldsymbol{v}_j$, then

$$\boldsymbol{x}(t) = ce^{\lambda_j t}\boldsymbol{v}_j,$$

so the solution remains on the same line through the origin.

5.3 Nodes

Suppose

$$\lambda_1 < 0, \qquad \lambda_2 < 0.$$

Then

$$e^{\lambda_1 t}, \quad e^{\lambda_2 t} \longrightarrow 0$$

as $t \to +\infty$. Hence every solution sufficiently near the origin approaches the origin.

Definition 5.3.1. In this case the origin is called a *stable node*, or *sink*.

If instead

$$\lambda_1 > 0, \qquad \lambda_2 > 0,$$

then nonzero solutions move away from the origin as $t \to +\infty$.

Definition 5.3.2. In this case the origin is called an *unstable node*, or *source*.

When

$$|\lambda_1| \neq |\lambda_2|,$$

one exponential mode dominates the other for large positive or negative time. Consequently, generic trajectories become tangent to one of the eigendirections.

5.4 Saddle Points

Suppose the two eigenvalues have opposite signs:

$$\lambda_1 < 0 < \lambda_2.$$

Then

$$\boldsymbol{x}(t) = c_1 e^{\lambda_1 t}\boldsymbol{v}_1 + c_2 e^{\lambda_2 t}\boldsymbol{v}_2.$$

Along the eigendirection corresponding to $\lambda_1 < 0$, solutions approach the origin as $t \to +\infty$. Along the eigendirection corresponding to $\lambda_2 > 0$, solutions move away from the origin.

Definition 5.4.1. An equilibrium with real eigenvalues of opposite signs is called a *saddle point*.

A saddle point is unstable.
The two eigendirections form the distinguished incoming and outgoing directions of the phase portrait.

5.5 Repeated Real Eigenvalues

Suppose

$$\lambda_1 = \lambda_2 = \lambda.$$

If A has two linearly independent eigenvectors, then

$$A = \lambda I$$

in a suitable eigenbasis, and

$$\boldsymbol{x}(t) = e^{\lambda t}\boldsymbol{x}_0.$$

Every line through the origin is a trajectory.
If $\lambda < 0$, the origin is a stable node. If $\lambda > 0$, it is an unstable node.
If there is only one independent eigenvector, a generalized eigenvector is required. The solution contains a term of the form

$$e^{\lambda t}\left(\boldsymbol{v}_2 + t\boldsymbol{v}_1\right).$$

The trajectories are no longer straight lines except along the true eigendirection.

Definition 5.5.1. This configuration is usually called an *improper node*.

Its stability is again determined by the sign of λ.

5.6 Complex Eigenvalues

Suppose

$$\lambda_{1,2} = \alpha \pm i\beta, \qquad \beta \neq 0.$$

As derived in Chapter 4, real solutions have the form

$$e^{\alpha t}\left[c_1\boldsymbol{u}(t) + c_2\boldsymbol{v}(t)\right],$$

where the vectors $\boldsymbol{u}(t)$ and $\boldsymbol{v}(t)$ are periodic combinations of $\cos\beta t$ and $\sin\beta t$.
If

$$\alpha < 0,$$

the factor $e^{\alpha t}$ tends to zero, so trajectories spiral toward the origin.

Definition 5.6.1. The origin is then a *stable spiral point* or *spiral sink*.

If

$$\alpha > 0,$$

the trajectories spiral away from the origin.

Definition 5.6.2. The origin is then an *unstable spiral point* or *spiral source*.

5.7 Centers

Suppose

$$\lambda_{1,2} = \pm i\beta, \qquad \beta \neq 0.$$

There is no exponential growth or decay. The trajectories are closed curves surrounding the origin.

Definition 5.7.1. The origin is called a *center*.

A center is stable, but not asymptotically stable: nearby solutions remain near the origin, but nonconstant solutions do not converge to the origin.

Remark 5.7.2. A center and a spiral can look similar over a short time interval, but their long-time behavior is fundamentally different. A spiral moves toward or away from the equilibrium, while a center remains on a closed orbit.

5.8 Classification by Trace and Determinant

For a matrix

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix},$$

the characteristic polynomial is

$$\lambda^2 - (\operatorname{tr} A)\lambda + \det A = 0.$$

Set

$$T = \operatorname{tr} A, \qquad D = \det A.$$

Then

$$\lambda_{1,2} = \frac{T \pm \sqrt{T^2 - 4D}}{2}.$$

Hence the phase portrait can be classified using T, D, and the discriminant

$$\Delta = T^2 - 4D.$$

If

$$D < 0,$$

the eigenvalues have opposite signs, so the origin is a saddle.
If

$$D > 0, \qquad \Delta > 0,$$

the eigenvalues are real and have the same sign. The origin is a node, stable when $T < 0$ and unstable when $T > 0$.
If

$$D > 0, \qquad \Delta < 0,$$

the eigenvalues are complex conjugates. The origin is a spiral when $T \neq 0$, and a center when $T = 0$.

5.9 Autonomous Systems

We now leave the linear setting and consider a planar nonlinear autonomous system

$$\begin{cases} x' = F(x, y), \\ y' = G(x, y). \end{cases}$$

Equivalently,

$$\boldsymbol{x}' = \boldsymbol{f}(\boldsymbol{x}),$$

where

$$\boldsymbol{f}(x, y) = \begin{pmatrix} F(x, y) \\ G(x, y) \end{pmatrix}.$$

The word *autonomous* means that the vector field does not depend explicitly on time.
If

$$\boldsymbol{x}(t)$$

is a solution, then for every constant t_0,

$$\boldsymbol{y}(t) = \boldsymbol{x}(t + t_0)$$

is also a solution wherever it is defined.
Indeed,

$$\boldsymbol{y}'(t) = \boldsymbol{x}'(t + t_0) = \boldsymbol{f}(\boldsymbol{x}(t + t_0)) = \boldsymbol{f}(\boldsymbol{y}(t)).$$

Thus time translation moves the starting point along the same trajectory without changing the trajectory itself.

5.10 Trajectories Do Not Cross

Assume the autonomous system satisfies the conditions for local uniqueness.
Suppose two trajectories intersect at a point $\boldsymbol{x}_0$. By shifting the time variables, both trajectories can be regarded as solutions of the same initial value problem

$$\boldsymbol{x}(0) = \boldsymbol{x}_0.$$

By uniqueness, the two solutions must coincide.

Proposition 5.10.1. Distinct trajectories of an autonomous system cannot intersect at a regular point when uniqueness holds.

Remark 5.10.2. This simple consequence of uniqueness is one of the reasons phase portraits are meaningful. The plane is organized into trajectories, rather than arbitrary curves that cross one another.

5.11 Critical Points

Definition 5.11.1. A point

$$(x_0, y_0)$$

is called a *critical point*, *equilibrium point*, or *stationary point* if

$$F(x_0, y_0) = 0, \qquad G(x_0, y_0) = 0.$$

The constant functions

$$x(t) \equiv x_0, \qquad y(t) \equiv y_0$$

then form a solution.
Thus critical points are precisely the constant trajectories of an autonomous system.

5.12 Stability of Critical Points

Let

$$\boldsymbol{x}_*$$

be an equilibrium of

$$\boldsymbol{x}' = \boldsymbol{f}(\boldsymbol{x}).$$

Definition 5.12.1. The equilibrium $\boldsymbol{x}_*$ is called *stable* if, for every $\varepsilon > 0$, there exists $\delta > 0$ such that

$$\|\boldsymbol{x}(0) - \boldsymbol{x}_*\| < \delta$$

implies

$$\|\boldsymbol{x}(t) - \boldsymbol{x}_*\| < \varepsilon$$

for all $t \geq 0$ for which the solution exists.

Thus stability means that a sufficiently small perturbation of the initial condition remains small for all future time.

Definition 5.12.2. The equilibrium $\boldsymbol{x}_*$ is called *asymptotically stable* if it is stable and there exists $\delta_0 > 0$ such that

$$\|\boldsymbol{x}(0) - \boldsymbol{x}_*\| < \delta_0$$

implies

$$\boldsymbol{x}(t) \longrightarrow \boldsymbol{x}_*$$

as $t \to +\infty$.

Definition 5.12.3. An equilibrium is *unstable* if it is not stable.

The distinction is important. A center is stable but not asymptotically stable, whereas a stable node or stable spiral is asymptotically stable.

5.13 Attraction and Stability

Stability and attraction describe different properties.
Stability asks whether nearby trajectories remain nearby:

$$\boldsymbol{x}(0) \approx \boldsymbol{x}_* \quad \Longrightarrow \quad \boldsymbol{x}(t) \approx \boldsymbol{x}_*$$

for all future time.
Attraction asks whether trajectories actually converge:

$$\boldsymbol{x}(t) \longrightarrow \boldsymbol{x}_*.$$

Asymptotic stability requires both.

5.14 Basin of Attraction

Definition 5.14.1. Suppose $\boldsymbol{x}_*$ is an asymptotically stable equilibrium. Its *basin of attraction*, or *region of attraction*, is the set of initial points whose forward solutions converge to $\boldsymbol{x}_*$:

$$\mathcal{B}(\boldsymbol{x}_*) = \left\{ \boldsymbol{x}_0 : \lim_{t \to +\infty} \boldsymbol{x}(t; \boldsymbol{x}_0) = \boldsymbol{x}_* \right\}.$$

Local asymptotic stability guarantees that the basin contains a neighborhood of the equilibrium, but the complete basin may be much larger.
Its boundary may be formed by special trajectories, such as those associated with saddle points.

5.15 Invariant Curves and Separating Trajectories

Certain trajectories divide the phase plane into regions with different long-time behavior. For a linear saddle, the eigendirections are invariant. One direction contains trajectories approaching the equilibrium, while the other contains trajectories leaving it.
In nonlinear systems, analogous special trajectories can act as boundaries between different regions of attraction.
The geometry of these curves is often more informative than an explicit solution formula.

5.16 Local Linearization

Consider

$$\begin{cases} x' = F(x,y), \\ y' = G(x,y), \end{cases}$$

and suppose

$$(x_0, y_0)$$

is a critical point.
Introduce

$$u = x - x_0, \qquad v = y - y_0.$$

Using the first-order Taylor expansion near (x_0, y_0),

$$F(x,y) = F(x_0,y_0) + F_x(x_0,y_0)u + F_y(x_0,y_0)v + \text{higher-order terms.}$$

Since

$$F(x_0, y_0) = 0,$$

we obtain

$$u' = F_x(x_0,y_0)u + F_y(x_0,y_0)v + \text{higher-order terms.}$$

Similarly,

$$v' = G_x(x_0,y_0)u + G_y(x_0,y_0)v + \text{higher-order terms.}$$

Therefore the local system has the form

$$\begin{pmatrix} u' \\ v' \end{pmatrix} = J(x_0,y_0)\begin{pmatrix} u \\ v \end{pmatrix} + \boldsymbol{g}(u,v),$$

where

$$\frac{\|\boldsymbol{g}(u,v)\|}{\sqrt{u^2+v^2}} \longrightarrow 0$$

as

$$(u,v) \longrightarrow (0,0).$$

5.17 The Jacobian Matrix

Definition 5.17.1. The matrix

$$J(x,y) = \begin{pmatrix} F_x(x,y) & F_y(x,y) \\ G_x(x,y) & G_y(x,y) \end{pmatrix}$$

is called the *Jacobian matrix* of the vector field.

At a critical point (x_0, y_0), the linearized system is

$$\begin{pmatrix} u' \\ v' \end{pmatrix} = J(x_0, y_0) \begin{pmatrix} u \\ v \end{pmatrix}.$$

Thus the nonlinear system near an equilibrium is approximated by a linear system, whose phase portrait can be classified using the eigenvalues of the Jacobian.

5.18 Isolated Critical Points

The Jacobian also gives a useful criterion for whether a critical point is isolated.

Proposition 5.18.1. Suppose

$$F(x_0, y_0) = 0, \qquad G(x_0, y_0) = 0,$$

and

$$\det J(x_0, y_0) \neq 0.$$

Then (x_0, y_0) is an isolated critical point.

This follows locally from the inverse function theorem applied to

$$(x, y) \longmapsto \Big(F(x, y), G(x, y)\Big).$$

5.19 Hyperbolic Critical Points

Definition 5.19.1. A critical point is called *hyperbolic* if no eigenvalue of its Jacobian matrix has zero real part.

For a hyperbolic critical point, the linearized system gives the correct local qualitative type.
In particular:

- if both eigenvalues have negative real parts, the critical point is locally asymptotically stable;
- if at least one eigenvalue has positive real part, the critical point is unstable;
- if the eigenvalues have opposite signs, the critical point has saddle-type local behavior.

Thus stable nodes, unstable nodes, saddles, stable spirals, and unstable spirals persist under sufficiently small nonlinear perturbations near the equilibrium.

5.20 When Linearization Is Inconclusive

Linearization does not always determine the behavior of a nonlinear system.
If an eigenvalue of the Jacobian has zero real part, the critical point is nonhyperbolic. Important examples include

$$\lambda = 0$$

and

$$\lambda_{1,2} = \pm i\beta.$$

In these cases, the higher-order nonlinear terms may change the qualitative behavior completely.
For example, the linearized system may have a center, while the nonlinear system may have a stable or unstable spiral.

Remark 5.20.1. Therefore one should not conclude that a nonlinear critical point is a center merely because its linearization has purely imaginary eigenvalues.

This limitation motivates a method that does not depend entirely on linearization: the Lyapunov second method.

5.21 Connection with the Previous Chapters

The passage from a nonlinear system to its linearization can be summarized as

$$\boldsymbol{x}' = \boldsymbol{f}(\boldsymbol{x}) \quad \longrightarrow \quad \boldsymbol{u}' = J(\boldsymbol{x}_*)\boldsymbol{u} + \text{higher-order terms.}$$

Chapter 4 developed a complete theory for the linear part

$$\boldsymbol{u}' = J(\boldsymbol{x}_*)\boldsymbol{u}.$$

The present chapter explains how its eigenvalues determine the local geometry of nonlinear trajectories whenever the equilibrium is hyperbolic.
When the linear approximation is insufficient, a different idea is needed. Instead of solving the differential equation or studying only its Jacobian, one may search for a scalar function that changes monotonically along trajectories. This leads to Lyapunov's second method, which is the starting point of the next chapter.

Chapter 6

Lyapunov Stability and Periodic Orbits

6.1 Lyapunov's Second Method

Consider the planar autonomous system

$$\begin{cases} x' = F(x, y), \\ y' = G(x, y), \end{cases}$$

and suppose that the origin is an isolated critical point:

$$F(0, 0) = G(0, 0) = 0.$$

Local linearization is often sufficient when the critical point is hyperbolic. However, when the eigenvalues of the Jacobian have zero real part, the linearized system may fail to determine the true behavior.
Lyapunov's second method studies stability without explicitly solving the differential equation. The basic idea is to construct a scalar function $V(x, y)$ that behaves like an energy and whose value changes monotonically along trajectories.

6.2 Positive and Negative Definite Functions

Definition 6.2.1. Let $D \subset \mathbb{R}^2$ be a neighborhood of the origin and let

$$V : D \longrightarrow \mathbb{R}$$

be continuous with $V(0, 0) = 0$.
The function V is called *positive definite* if

$$V(x, y) > 0$$

for every

$$(x, y) \in D \setminus \{(0, 0)\}.$$

It is called *positive semidefinite* if

$$V(x, y) \geq 0$$

throughout D.
Similarly, V is *negative definite* if

$$V(x, y) < 0$$

for every nonzero point in D, and *negative semidefinite* if

$$V(x, y) \leq 0.$$

A typical positive definite function is

$$V(x, y) = x^2 + y^2.$$

More generally, consider the quadratic form

$$V(x, y) = ax^2 + 2bxy + cy^2.$$

Its matrix representation is

$$V(\boldsymbol{x}) = \boldsymbol{x}^T \begin{pmatrix} a & b \\ b & c \end{pmatrix} \boldsymbol{x}.$$

For a real symmetric 2×2 matrix, the quadratic form is positive definite when

$$a > 0, \qquad ac - b^2 > 0.$$

6.3 Derivative Along a Trajectory

Suppose $V \in C^1(D)$. Along a solution

$$(x(t), y(t)),$$

the chain rule gives

$$\frac{d}{dt}V(x(t), y(t)) = V_x(x(t), y(t))x'(t) + V_y(x(t), y(t))y'(t).$$

Since

$$x' = F(x, y), \qquad y' = G(x, y),$$

we obtain

$$\dot{V}(x, y) = V_x(x, y)F(x, y) + V_y(x, y)G(x, y).$$

In vector notation,

$$\dot{V} = \nabla V \cdot \boldsymbol{f}.$$

The sign of $\dot{V}$ describes whether V increases or decreases along trajectories.

6.4 Lyapunov Stability Theorem

Theorem 6.4.1 (Lyapunov Stability Theorem)**.** Suppose the origin is an isolated critical point and $V \in C^1(D)$ satisfies

$$V(0, 0) = 0.$$

If V is positive definite and

$$\dot{V} \leq 0$$

in D, then the origin is stable.
If V is positive definite and

$$\dot{V} < 0$$

for every

$$(x, y) \neq (0, 0),$$

then the origin is asymptotically stable.

Proof. Assume first that V is positive definite and $\dot{V} \leq 0$.
Choose $\varepsilon > 0$ sufficiently small so that the closed disk

$$\overline{B_\varepsilon(0)}$$

lies inside D. Since V is positive on the circle

$$x^2 + y^2 = \varepsilon^2,$$

compactness gives

$$m_\varepsilon = \min_{x^2+y^2=\varepsilon^2} V(x, y) > 0.$$

By continuity of V at the origin, there exists $\delta > 0$ such that

$$x^2 + y^2 < \delta^2$$

implies

$$V(x, y) < m_\varepsilon.$$

Along any trajectory,

$$\dot{V} \leq 0,$$

so

$$V(x(t), y(t)) \leq V(x(0), y(0)) < m_\varepsilon.$$

Therefore the trajectory cannot reach the circle

$$x^2 + y^2 = \varepsilon^2,$$

because $V \geq m_\varepsilon$ there. Hence the origin is stable.
If in addition $\dot{V} < 0$ away from the origin, the value of V strictly decreases along every nonconstant trajectory. This forces nearby trajectories toward the equilibrium, giving asymptotic stability. □

6.5 Geometric Meaning of a Lyapunov Function

If V is positive definite, the level curves

$$V(x, y) = c$$

surround the origin for sufficiently small $c > 0$.
If

$$\dot{V} < 0,$$

then a trajectory crosses these level curves toward smaller values of V. Thus the motion is directed inward toward the equilibrium.
If

$$\dot{V} = 0,$$

the trajectory remains on the same level curve.
This geometric interpretation explains why V is often regarded as an energy-like function.

6.6 Lyapunov Instability Criterion

Lyapunov functions can also be used to prove instability.

Theorem 6.6.1 (Lyapunov Instability Theorem)**.** Suppose

$$V(0, 0) = 0$$

and every neighborhood of the origin contains points at which

$$V(x, y) > 0.$$

Assume that there is a region where $V > 0$ and

$$\dot{V} > 0.$$

Then the origin is unstable.

The idea is that once a trajectory enters a region in which $V > 0$ and $\dot{V} > 0$, the value of V increases rather than returning toward zero. Hence arbitrarily small perturbations may eventually move away from the equilibrium.

6.7 Constructing Lyapunov Functions

There is no universal procedure for finding a Lyapunov function.
For many systems, a useful first attempt is a quadratic function

$$V(x,y) = ax^2 + 2bxy + cy^2.$$

Its derivative along trajectories is

$$\dot{V} = (2ax + 2by)F(x,y) + (2bx + 2cy)G(x,y).$$

The coefficients a, b, c may then be chosen so that V is positive definite while $\dot{V}$ has the desired sign.

Remark 6.7.1. Finding a Lyapunov function is often the difficult part of the method. Once such a function is found, the stability argument itself is usually short.

6.8 A Quadratic Lyapunov Example

Consider

$$\begin{cases} x' = -x - y, \\ y' = x - y. \end{cases}$$

Take

$$V(x,y) = x^2 + y^2.$$

Then V is positive definite. Moreover,

$$\begin{aligned} \dot{V} &= 2xx' + 2yy' \\ &= 2x(-x - y) + 2y(x - y) \\ &= -2x^2 - 2y^2. \end{aligned}$$

Thus

$$\dot{V} < 0$$

whenever $(x,y) \neq (0,0)$.
Therefore the origin is asymptotically stable.

6.9 First Integrals

There is another important possibility:

$$\dot{V} = 0.$$

Definition 6.9.1. A nonconstant function $H(x,y)$ is called a *first integral* or *constant of motion* of the system if

$$\frac{d}{dt} H(x(t), y(t)) = 0$$

along every solution.

Hence

$$H(x(t), y(t)) = C$$

along each trajectory.
The trajectories therefore lie on level curves

$$H(x,y) = C.$$

Remark 6.9.2. A Lyapunov function decreases or increases along trajectories, whereas a first integral remains constant. These two situations correspond to dissipative and conservative behavior, respectively.

6.10 Hamiltonian-Type Systems

Suppose the system has the form

$$\begin{cases} x' = H_y(x, y), \\ y' = -H_x(x, y). \end{cases}$$

Then

$$\begin{aligned} \dot{H} &= H_x x' + H_y y' \\ &= H_x H_y - H_y H_x \\ &= 0. \end{aligned}$$

Thus H is constant along trajectories.
Such systems provide an important source of closed trajectories and periodic solutions.

6.11 The Lotka–Volterra First Integral

Recall the predator–prey system

$$\begin{cases} x' = x(r - by), \\ y' = y(-s + cx), \end{cases}$$

where

$$r, b, s, c > 0.$$

From Chapter 1, the function

$$H(x, y) = s \ln x - cx + r \ln y - by$$

is constant along positive trajectories.
Indeed,

$$\dot{H} = 0.$$

The positive equilibrium is

$$\left(\frac{s}{c}, \frac{r}{b} \right).$$

The level curves of H surrounding this point describe the closed orbits of the system.
This is an example in which the geometry of trajectories can be found without solving explicitly for $x(t)$ and $y(t)$.

6.12 A Conservative Second-Order Equation

Consider

$$x'' + f(x) = 0,$$

where f is continuous.
Introduce

$$y = x'.$$

Then

$$\begin{cases} x' = y, \\ y' = -f(x). \end{cases}$$

Let F be an antiderivative of f:

$$F'(x) = f(x).$$

Define

$$H(x, y) = \frac{1}{2}y^2 + F(x).$$

Along a trajectory,

$$\begin{aligned}\dot{H} &= yy' + F'(x)x' \\ &= y(-f(x)) + f(x)y \\ &= 0.\end{aligned}$$

Thus

$$\frac{1}{2}y^2 + F(x) = C.$$

This is the energy conservation law for the second-order equation.

6.13 Stability from the Potential Energy

Suppose

$$f(x_0) = 0.$$

Then

$$(x_0, 0)$$

is an equilibrium.
Since

$$F'(x_0) = f(x_0) = 0,$$

the character of F near x_0 determines the geometry of the energy levels.
If F has a strict local minimum at x_0, then

$$H(x, y) - H(x_0, 0)$$

is positive definite near $(x_0, 0)$. Because

$$\dot{H} = 0,$$

the nearby energy curves remain close to the equilibrium. Hence

$$(x_0, 0)$$

is stable.
If F has a strict local maximum at x_0, the equilibrium has saddle-type behavior and is unstable.

6.14 Periodic Solutions and Closed Trajectories

Consider again

$$\boldsymbol{x}' = \boldsymbol{f}(\boldsymbol{x})$$

in the plane.

Definition 6.14.1. A nonconstant solution $\boldsymbol{x}(t)$ is called *periodic* if there exists $T > 0$ such that

$$\boldsymbol{x}(t + T) = \boldsymbol{x}(t)$$

for every t.

The smallest such positive T, when it exists, is called the *period*.
The trajectory traced out by a periodic solution is a closed curve in the phase plane.
Conversely, because solutions of an autonomous system are unique, a nonconstant closed trajectory corresponds to a periodic solution.

6.15 Periodic Orbits

Definition 6.15.1. The closed trajectory corresponding to a periodic solution is called a *periodic orbit*.

The distinction between a periodic solution and a periodic orbit is important.
A periodic solution

$$\boldsymbol{x}(t)$$

contains a particular parametrization in time, whereas the orbit is the geometric curve

$$\{\boldsymbol{x}(t) : t \in \mathbb{R}\}.$$

Different time shifts

$$\boldsymbol{x}(t + t_0)$$

represent the same periodic orbit.

6.16 Orbital Stability

For an equilibrium point, stability means that nearby solutions remain close to the same point. For a periodic orbit, this definition must be modified because two nearby solutions may move around the same closed curve with different phases.

Definition 6.16.1. A periodic orbit Γ is called *orbitally stable* if trajectories starting sufficiently close to Γ remain close to the set Γ for all future time.

It is called *asymptotically orbitally stable* if, in addition, the distance from nearby trajectories to Γ tends to zero as $t \to +\infty$.
Thus orbital stability concerns distance to the entire orbit, not distance to one specified moving point on the orbit.

6.17 Limit Cycles

Definition 6.17.1. A periodic orbit is called a *limit cycle* if it is isolated from other periodic orbits.

A stable limit cycle attracts nearby trajectories, whereas an unstable limit cycle repels nearby trajectories.
There may also be a semistable limit cycle, attracting trajectories from one side and repelling them from the other.

Remark 6.17.2. A center is therefore fundamentally different from a limit cycle. Near a center there is usually a continuous family of closed trajectories, so none of them is isolated.

6.18 Limit Sets

To describe the long-time behavior of a trajectory, it is useful to record the points that it approaches repeatedly.

Definition 6.18.1. Let $\boldsymbol{x}(t)$ be a solution defined for all sufficiently large t. Its *ω-limit set* is

$$\omega(\boldsymbol{x}) = \left\{ \boldsymbol{p} : \begin{array}{l} \text{there exists } t_n \to +\infty \\ \text{such that } \boldsymbol{x}(t_n) \to \boldsymbol{p} \end{array} \right\}.$$

Thus the ω-limit set contains all accumulation points of the trajectory as $t \to +\infty$.
Possible limiting behavior in the plane includes an equilibrium point or a periodic orbit.
More complicated configurations may occur when critical points and connecting trajectories are present.

6.19 Positively Invariant Regions

Definition 6.19.1. A region R is called *positively invariant* if every solution starting in R remains in R for all future time.

A compact positively invariant region acts as a trapping region. Trajectories can enter such a region and remain there indefinitely.
This idea is central to proving the existence of periodic orbits.

6.20 Poincaré–Bendixson Theorem

The most important qualitative theorem for planar autonomous systems describes the possible limiting behavior of bounded trajectories.

Theorem 6.20.1 (Poincaré–Bendixson)**.** Let $D \subset \mathbb{R}^2$ be a domain, and suppose a forward trajectory remains in a compact subset of D.
If its ω-limit set contains no critical point, then the ω-limit set is a periodic orbit.

This theorem is special to two-dimensional autonomous systems.
It gives a mechanism for proving that a periodic solution exists even when the differential equation cannot be solved explicitly.

6.21 A Trapping-Region Criterion

A particularly useful consequence is the following.

Corollary 6.21.1. Let $R \subset D$ be a compact positively invariant region. If R contains no critical point, then R contains a periodic orbit.

Indeed, choose any trajectory starting in R. Since R is compact and positively invariant, the forward trajectory remains bounded. Its ω-limit set is nonempty and remains in R.
Because there are no critical points in R, the Poincaré–Bendixson theorem implies that the limit set is a periodic orbit.

6.22 Critical Points Inside a Periodic Orbit

A periodic orbit separates the plane into an interior and an exterior.
The course notes use the following geometric restriction.

Theorem 6.22.1. Let Γ be a closed orbit of a planar autonomous system. Then the interior of Γ contains at least one critical point.

In particular, if the interior contains exactly one isolated critical point, that point cannot have saddle-type geometry.
This gives a useful way to rule out certain proposed phase portraits.

6.23 Bendixson's Criterion

The Poincaré–Bendixson theorem can prove the existence of periodic orbits. Bendixson's criterion often proves their nonexistence.
Consider

$$\begin{cases} x' = F(x, y), \\ y' = G(x, y), \end{cases}$$

with

$$F, G \in C^1(D).$$

Theorem 6.23.1 (Bendixson Criterion)**.** Suppose $D \subset \mathbb{R}^2$ is simply connected and

$$F_x + G_y$$

does not change sign in D and is not identically zero.
Then there is no periodic orbit lying entirely inside D.

Proof. Suppose, to the contrary, that a periodic orbit Γ lies inside D, and let R denote its interior.
Along the trajectory,

$$dx = F\,dt, \qquad dy = G\,dt.$$

Hence

$$F\,dy - G\,dx = (FG - GF)\,dt = 0.$$

Therefore

$$\oint_\Gamma F\,dy - G\,dx = 0.$$

By Green's theorem,

$$\oint_\Gamma F\,dy - G\,dx = \iint_R (F_x + G_y)\,dx\,dy.$$

If

$$F_x + G_y$$

has one strict sign throughout R, the double integral cannot be zero, giving a contradiction. □

Remark 6.23.2. The quantity

$$F_x + G_y = \nabla \cdot \boldsymbol{f}$$

is the divergence of the vector field.
Thus a vector field that continuously expands area or continuously contracts area throughout a simply connected region cannot contain a closed orbit entirely within that region.

6.24 Existence versus Nonexistence of Periodic Orbits

The two principal tools developed above work in opposite directions.
The Poincaré–Bendixson theorem has the form

$$\begin{array}{c} \text{bounded forward trajectory} \\ + \\ \text{no critical point in the limit set} \end{array} \quad \Longrightarrow \quad \text{periodic orbit.}$$

Bendixson's criterion has the form

$$\nabla \cdot \boldsymbol{f} \text{ has one sign} \quad \Longrightarrow \quad \text{no periodic orbit.}$$

Together they provide a qualitative theory of periodic behavior that requires no explicit solution formula.

6.25 Summary of the Dynamical Viewpoint

For nonlinear autonomous systems, the central problem is often no longer to compute

$$\boldsymbol{x}(t)$$

explicitly. Instead, one asks:

- where are the equilibrium points;
- which equilibria are stable;
- whether trajectories are trapped;
- whether periodic orbits exist;
- whether those periodic orbits attract or repel nearby trajectories;
- what the possible long-time limit sets are.

Local linearization answers many of these questions near hyperbolic critical points. Lyapunov functions provide information when linearization is insufficient. First integrals identify conservative geometry, while the Poincaré–Bendixson and Bendixson theorems control periodic behavior in the plane.

This completes the qualitative dynamical part of the course. The final chapter changes viewpoint again: instead of initial value problems and long-time dynamics, we study boundary value problems and the eigenvalue structure of differential operators.

Chapter 7

Sturm–Liouville Theory and Boundary Value Problems

7.1 From Initial Values to Boundary Values

Most differential equations studied so far have been initial value problems. For example,

$$y'' + p(x)y' + q(x)y = g(x)$$

is accompanied by conditions such as

$$y(x_0) = y_0, \qquad y'(x_0) = y_1.$$

A boundary value problem is different. Instead of prescribing all information at one point, conditions are imposed at different endpoints of an interval.
For example,

$$y'' + \lambda y = 0, \qquad a < x < b,$$

may be accompanied by

$$y(a) = 0, \qquad y(b) = 0.$$

Unlike an ordinary initial value problem, a boundary value problem need not have a solution for every value of the parameter λ. Nontrivial solutions may exist only for special values of λ.
These special values are called eigenvalues.

7.2 Boundary Conditions

Consider a second-order differential equation on

$$a \le x \le b.$$

Several standard boundary conditions appear frequently.

Definition 7.2.1. A *Dirichlet boundary condition* prescribes the values of the function at the endpoints, for example

$$y(a) = 0, \qquad y(b) = 0.$$

Definition 7.2.2. A *Neumann boundary condition* prescribes the derivatives:

$$y'(a) = 0, \qquad y'(b) = 0.$$

More generally, separated linear boundary conditions have the form

$$\alpha_1 y(a) + \alpha_2 y'(a) = 0,$$

and

$$\beta_1 y(b) + \beta_2 y'(b) = 0,$$

where

$$\alpha_1^2 + \alpha_2^2 > 0, \qquad \beta_1^2 + \beta_2^2 > 0.$$

Conditions involving both y and y' are often called *Robin boundary conditions.*

7.3 A Basic Eigenvalue Problem

Consider

$$y'' + \lambda y = 0$$

with homogeneous Dirichlet boundary conditions

$$y(0) = 0, \qquad y(\pi) = 0.$$

We separate the discussion according to the sign of λ.
If

$$\lambda = -\mu^2 < 0,$$

then

$$y'' - \mu^2 y = 0,$$

so

$$y(x) = c_1 e^{\mu x} + c_2 e^{-\mu x}.$$

The boundary conditions force

$$c_1 = c_2 = 0.$$

Hence there is no nontrivial solution.
If

$$\lambda = 0,$$

then

$$y'' = 0,$$

so

$$y(x) = c_1 x + c_2.$$

Again the boundary conditions give only the trivial solution.
Finally, suppose

$$\lambda = \mu^2 > 0.$$

Then

$$y(x) = c_1 \cos \mu x + c_2 \sin \mu x.$$

The condition $y(0) = 0$ gives $c_1 = 0$, and

$$y(\pi) = 0$$

requires

$$\sin(\mu\pi) = 0.$$

Hence

$$\mu = n, \qquad n = 1, 2, \ldots.$$

Therefore the eigenvalues are

$$\lambda_n = n^2,$$

with corresponding eigenfunctions

$$y_n(x) = \sin(nx).$$

Remark 7.3.1. This example already shows the essential phenomenon of a Sturm–Liouville problem: the parameter λ cannot be chosen arbitrarily. Nontrivial solutions occur only for a discrete collection of eigenvalues.

7.4 Sturm–Liouville Problems

The general Sturm–Liouville equation considered in this course has the form

$$-\frac{d}{dx}\left(p(x)y'(x)\right) + q(x)y(x) = \lambda r(x)y(x), \qquad a < x < b.$$

Equivalently,

$$-(p(x)y')' + q(x)y = \lambda r(x)y.$$

The boundary conditions are

$$\alpha_1 y(a) + \alpha_2 y'(a) = 0,$$

and

$$\beta_1 y(b) + \beta_2 y'(b) = 0.$$

For a regular Sturm–Liouville problem, the coefficient functions satisfy the usual regularity assumptions, with

$$p(x) > 0, \qquad r(x) > 0$$

on the interval.
The function $r(x)$ is called the *weight function*.

7.5 Differential Operator Form

Define the differential operator

$$L(y) = -(p(x)y')' + q(x)y.$$

Then the Sturm–Liouville equation becomes

$$L(y) = \lambda r(x)y.$$

This form makes the analogy with finite-dimensional linear algebra clear:

$$A\boldsymbol{v} = \lambda \boldsymbol{v}$$

is replaced by

$$L(y) = \lambda r(x)y.$$

The unknown is now a function rather than a vector, but the concepts of eigenvalue and eigenfunction remain analogous.

7.6 Eigenvalues and Eigenfunctions

Definition 7.6.1. A number λ is called an *eigenvalue* of the Sturm–Liouville problem if there exists a nonzero function y satisfying

$$L(y) = \lambda r(x)y$$

together with the prescribed boundary conditions.

Definition 7.6.2. A corresponding nonzero solution y is called an *eigenfunction* associated with λ.

As usual, any nonzero scalar multiple of an eigenfunction is another eigenfunction corresponding to the same eigenvalue.

7.7 Weighted Inner Product

Because the eigenvalue equation contains the weight $r(x)$, the natural inner product is weighted.

Definition 7.7.1. For suitable functions u, v, define

$$\langle u, v \rangle_r = \int_a^b r(x)u(x)v(x)\,dx.$$

When $r(x) > 0$,

$$\|u\|_r^2 = \langle u, u \rangle_r = \int_a^b r(x)u(x)^2\,dx.$$

Two functions are called orthogonal with respect to the weight r if

$$\langle u, v \rangle_r = 0.$$

7.8 Lagrange's Identity

The algebraic foundation of Sturm–Liouville theory is an integration-by-parts identity. Let

$$L(u) = -(pu')' + qu$$

and

$$L(v) = -(pv')' + qv.$$

Then

$$uL(v) - vL(u) = -u(pv')' + v(pu')'.$$

A direct calculation gives

$$uL(v) - vL(u) = \frac{d}{dx}\left[p(x)\Big(vu' - uv'\Big)\right].$$

Therefore

$$\int_a^b \left[uL(v) - vL(u)\right] dx = \Big[p(x)\Big(vu' - uv'\Big)\Big]_a^b.$$

This is commonly called *Lagrange's identity*.

7.9 The Boundary Term

Suppose u and v satisfy the same separated homogeneous boundary conditions:

$$\alpha_1 u(a) + \alpha_2 u'(a) = 0,$$

$$\alpha_1 v(a) + \alpha_2 v'(a) = 0,$$

and similarly at b.
Then

$$u(a)v'(a) - u'(a)v(a) = 0,$$

and

$$u(b)v'(b) - u'(b)v(b) = 0.$$

Therefore

$$\Big[p(x)\Big(vu' - uv'\Big)\Big]_a^b = 0.$$

Hence

$$\int_a^b uL(v)\,dx = \int_a^b vL(u)\,dx.$$

7.10 Self-Adjointness

Definition 7.10.1. The differential operator L is called *self-adjoint* with respect to the boundary conditions if

$$\int_a^b uL(v)\,dx = \int_a^b L(u)v\,dx$$

for all admissible functions u and v.

For the Sturm–Liouville operator

$$L(y) = -(py')' + qy,$$

the separated homogeneous boundary conditions eliminate the boundary term in Lagrange's identity. Thus L is self-adjoint on the class of functions satisfying those conditions.

Remark 7.10.2. Self-adjointness is the structural reason behind the two most important properties of the Sturm–Liouville problem:

$$\text{real eigenvalues} \qquad \text{and} \qquad \text{orthogonal eigenfunctions.}$$

7.11 Reality of the Eigenvalues

Theorem 7.11.1. Every eigenvalue of a regular Sturm–Liouville problem is real.

Proof. Let y be an eigenfunction with eigenvalue λ:

$$L(y) = \lambda r y.$$

Allow y and λ temporarily to be complex. Taking complex conjugates gives

$$L(\overline{y}) = \overline{\lambda}\, r \overline{y},$$

since the coefficients are real.
Using self-adjointness,

$$\int_a^b \overline{y} L(y)\,dx = \int_a^b y L(\overline{y})\,dx.$$

Hence

$$\lambda \int_a^b r(x)\,|y(x)|^2\,dx = \overline{\lambda} \int_a^b r(x)\,|y(x)|^2\,dx.$$

Since y is nonzero and $r(x) > 0$,

$$\int_a^b r(x)\,|y(x)|^2\,dx > 0.$$

Therefore

$$\lambda = \overline{\lambda},$$

so $\lambda \in \mathbb{R}$. □

7.12 Simplicity of the Eigenvalues

Theorem 7.12.1. Each eigenvalue of a regular Sturm–Liouville problem with separated boundary conditions has a one-dimensional eigenspace.

Proof. Let u and v be two eigenfunctions corresponding to the same eigenvalue λ. Then

$$L(u) = \lambda r u,$$

and

$$L(v) = \lambda r v.$$

Lagrange's identity gives

$$\frac{d}{dx}\left[p(x)\Big(vu' - uv'\Big)\right] = 0.$$

Thus

$$p(x)\Big(vu' - uv'\Big) = C.$$

The boundary conditions imply $C = 0$, so

$$vu' - uv' = 0.$$

Hence the Wronskian of u and v vanishes. Therefore the two solutions are linearly dependent.
Thus every eigenfunction associated with λ is a scalar multiple of one fixed eigenfunction.
□

7.13 Orthogonality of Eigenfunctions

Theorem 7.13.1 (Orthogonality)**.** Suppose y_1 and y_2 are eigenfunctions corresponding to distinct eigenvalues

$$\lambda_1 \neq \lambda_2.$$

Then

$$\int_a^b r(x) y_1(x) y_2(x)\, dx = 0.$$

Proof. Since

$$L(y_1) = \lambda_1 r y_1$$

and

$$L(y_2) = \lambda_2 r y_2,$$

self-adjointness gives

$$\int_a^b y_1 L(y_2)\, dx = \int_a^b y_2 L(y_1)\, dx.$$

Therefore

$$\lambda_2 \int_a^b r(x) y_1 y_2\, dx = \lambda_1 \int_a^b r(x) y_1 y_2\, dx.$$

Hence

$$(\lambda_2 - \lambda_1) \int_a^b r(x) y_1 y_2\, dx = 0.$$

Since

$$\lambda_1 \neq \lambda_2,$$

we obtain

$$\int_a^b r(x) y_1(x) y_2(x)\, dx = 0.$$

□

Thus eigenfunctions corresponding to distinct eigenvalues are orthogonal with respect to the weighted inner product.

7.14 Normalization

An eigenfunction can be multiplied by any nonzero constant. Therefore it is often convenient to normalize it so that

$$\|y_n\|_r = 1.$$

That is,

$$\int_a^b r(x) y_n(x)^2 \, dx = 1.$$

After normalization, eigenfunctions associated with different eigenvalues satisfy

$$\langle y_m, y_n \rangle_r = \begin{cases} 1, & m = n, \\ 0, & m \neq n. \end{cases}$$

Hence the normalized eigenfunctions form an orthonormal family with respect to the weighted inner product.

7.15 Ordering of the Eigenvalues

For a regular Sturm–Liouville problem, the eigenvalues can be arranged in an increasing sequence

$$\lambda_1 < \lambda_2 < \lambda_3 < \cdots,$$

with

$$\lambda_n \longrightarrow +\infty$$

as $n \to \infty$.
Thus the spectrum is discrete rather than continuous.
Each eigenvalue has an eigenfunction

$$y_n(x),$$

which may be normalized so that

$$\int_a^b r(x) y_n(x)^2 \, dx = 1.$$

7.16 Zeros of Eigenfunctions

The oscillatory behavior of the eigenfunctions becomes increasingly complicated as the eigenvalue increases.

Theorem 7.16.1. Let y_n be an eigenfunction associated with the n-th eigenvalue λ_n. Then y_n changes sign $n - 1$ times in the open interval

$$(a, b).$$

Thus the first eigenfunction has no sign change in the interior, the second changes sign once, and so on.
This provides a geometric interpretation of the ordering of the eigenvalues: higher eigenvalues correspond to increasingly oscillatory eigenfunctions.

7.17 Eigenfunction Expansions

The orthogonality of the eigenfunctions suggests expanding functions in the form

$$f(x) \sim \sum_{n=1}^{\infty} c_n y_n(x).$$

Formally taking the weighted inner product with y_m gives

$$\langle f, y_m \rangle_r = \sum_{n=1}^{\infty} c_n \langle y_n, y_m \rangle_r .$$

If the eigenfunctions are normalized, orthogonality gives

$$c_m = \langle f, y_m \rangle_r .$$

Therefore

$$c_m = \int_a^b r(x) f(x) y_m(x)\, dx.$$

Without normalization,

$$c_m = \frac{\displaystyle\int_a^b r(x) f(x) y_m(x)\, dx}{\displaystyle\int_a^b r(x) y_m(x)^2\, dx}.$$

7.18 Connection with Fourier Series

The elementary problem

$$-y'' = \lambda y, \qquad y(0) = y(\pi) = 0,$$

has eigenfunctions

$$\sin x, \quad \sin 2x, \quad \sin 3x, \quad \ldots.$$

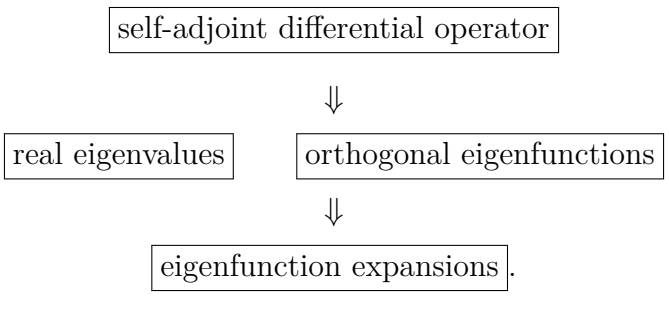
Hence its eigenfunction expansion is precisely a Fourier sine series:

$$f(x) \sim \sum_{n=1}^{\infty} c_n \sin(nx).$$

Sturm–Liouville theory therefore generalizes the familiar Fourier series idea. Instead of using trigonometric functions fixed in advance, the differential operator and boundary conditions determine the appropriate orthogonal family of functions.

7.19 Why Self-Adjointness Matters

The main structure of the theory can be summarized as



This is the infinite-dimensional analogue of the spectral behavior of real symmetric matrices.
For a real symmetric matrix A,

$$A\boldsymbol{v}_n = \lambda_n \boldsymbol{v}_n,$$

the eigenvalues are real and eigenvectors belonging to distinct eigenvalues are orthogonal.
For a Sturm–Liouville operator,

$$L(y_n) = \lambda_n r y_n,$$

the corresponding conclusions are

$$\lambda_n \in \mathbb{R}$$

and

$$\langle y_m, y_n \rangle_r = 0$$

whenever $m \neq n$.

7.20 Final Perspective

The development of ordinary differential equations in these notes can now be viewed as a sequence of changes in perspective.
For first-order equations, the initial goal was explicit computation:

$$\text{ODE} \longrightarrow \text{solution formula.}$$

Existence and uniqueness theory then showed that one can study a solution even when no formula is available:

$$\text{differential equation} \longrightarrow \text{existence, uniqueness, and extension.}$$

Linear equations introduced a vector-space structure:

$$\text{solutions} \longrightarrow \text{basis and Wronskian.}$$

Linear systems replaced scalar functions by vectors and matrices:

$$\text{linear ODE} \longrightarrow \text{eigenvalues and matrix exponentials.}$$

Nonlinear autonomous systems shifted the emphasis from formulas to geometry:

$$\text{solution curves} \longrightarrow \text{phase portraits and stability.}$$

Lyapunov functions and the Poincaré–Bendixson theory then provided tools for understanding long-time behavior without solving the system explicitly.
Finally, Sturm–Liouville theory changes the question once more. The central object is no longer one initial-value solution, but the spectral structure of a differential operator:

$$L(y) = \lambda r y.$$

The resulting eigenvalues and eigenfunctions connect ordinary differential equations with Fourier analysis and provide one of the basic mathematical structures underlying boundary value problems and partial differential equations.

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